The Gribov Process Through the CFAC Lens:

Reproducing Janssen–Täuber's 2-Loop DP Result
by Reduction to a Minimal Set of Master Integrals

Counting is fully discharged by Lagrange inversion; three master integrals — already evaluated by Janssen and Täuber — remain to be quoted

S. Amarteifio
Percolation Labs

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Abstract

This paper takes up G. Pruessner's challenge: to demonstrate, on a process *without* the rapidity-reversal-driven Ward identities of the branching Wiener sausage / BARW-even, that CFAC reduces the labour of a 2-loop renormalisation calculation to the smallest irreducible set of quantities that must be evaluated by hand. The test is the Gribov process (directed percolation) at $d_c = 4$.

Theorem (AC reduction of 2-loop DP). Take as input three *integrals* — the bubble B_2, the 2-loop sunset B_3^{sun}, and one 2-loop vertex master B_V — evaluated at the Janssen–Täuber 2005 (JT05) [3] symmetric Reggeon renormalisation point. (The output of any such evaluation, by hand, by Panzer's HyperInt [10], by Borinsky's tropical Monte Carlo [15], or quoted from JT05, is a Laurent expansion in ε.) Then *every other rational quantity* in the 2-loop renormalisation of DP — diagram counts, symmetry factors, vertex signs, β-function coefficients, Z-factor 1-loop pole rationals, Z-factor *2-loop double pole* rationals, fixed-point shifts, anomalous-dimension series, exponents — is determined by closed-form analytic-combinatorics operations:

(C1) Lagrange inversion on $G(z) = z(1 + G^2)$ produces all diagram counts c_Γ and symmetry factors automatically (Flajolet–Sedgewick A.6).

(C2) Bivariate marking $G(z, \alpha) = z(1 + \alpha G^2)$ evaluated at $\alpha = -1$ produces the rapidity-reversal sign assignment for each topology.

(C3) Kirchhoff's matrix-tree theorem produces every Symanzik polynomial U_Γ from graph structure.

(C4) Polynomial-derivative operators at the JT05 symmetric point on $U_{\mathsf{bubble}} = x_1 + x_2$ produce the four 1-loop algebra factors $a_1^X = \{\frac{1}{4}, \frac{1}{8}, \frac{1}{2}, 2\}$ (matching the JT05 minimal-subtraction choice between Eqs. (55)–(56)).

(C5) The MS-bar pole-cancellation identity (a.k.a. Connes–Kreimer Hopf-antipode evaluation)

$$\boxed{Z_X^{(2,2)} \;=\; \tfrac{1}{2}\, a_X^{(1)}\big(\beta_1 + a_X^{(1)}\big), \qquad \beta_1 = a_u^{(1)} - 2\, a_\psi^{(1)} = \tfrac{3}{2}}$$

produces all *four* 2-loop double poles $\{Z_\psi, Z_\lambda, Z_\tau, Z_u\}^{(2,2)} = \{\frac{7}{32}, \frac{13}{128}, \frac{1}{2}, \frac{7}{2}\}$, matching JT05 Eq. (57) exactly.

(C6) The 2-loop simple-pole BPHZ counterterm has the parallel closed form $\mathrm{BPHZ}_X = \frac{1}{2}\, a_X^{(1)}(a_X^{(1)} - \beta_1)$, also from the Hopf antipode and free of integral input.

(C7) The remaining (primitive) simple-pole content of Z_X decomposes via the counting–bridge–algebra factorisation through the 12 rational IBP coefficients $q_\bullet^X$, fixed by 5 structural CFAC constraints + the JT05 closure on the master residues. The three master integral values are the only places where integration enters.

(C8) The downstream pipeline $Z \to \gamma \to \beta \to u^* \to$ exponents is pure series algebra and reproduces JT05 Eqs. (58), (59), (60) with zero residual (SymPy-verified).

What this saves. In the direct JT05 calculation one enumerates ~ 10 self-energy + vertex graphs, assigns symmetry factors by hand, computes each Symanzik polynomial separately, and extracts BPHZ subtractions per graph. After CFAC:

	Direct JT05	**CFAC + IBP**
1PI graphs to enumerate	~ 10	0 (Lagrange inversion)
Symmetry factors to assign	~ 10	0 (Lagrange overcount)
Distinct loop integrals	$\sim$ 8–10	**3 master integrals**
System-specific rerun cost	full repeat	c_Γ recount only

Factoring out the counting. CFAC's contribution is to isolate *which* integrals are irreducible after the counting problem has been factored out, and to discharge everything else combinatorially. The remaining (small) set of master integrals either has well-known published values [3, 4] or admits further combinatorial reduction by Panzer's HyperInt [10], Borinsky's tropical Monte Carlo [15], and the cosmic-Galois selection rules [13].

Reading guide. Boxes: ■ gray = JT05 canonical equations; ■ green = analytic-combinatorics theorems we invoke; ■ orange = CFAC closed-form expressions that produce the diagrams directly from $G = z(1+G^2)$; ■ purple = statements mechanised in the accompanying SymPy script.

On the purple boxes. A purple box does not mean "trust the black box." Every purple-box computation is analytical: rational linear algebra, polynomial coefficient extraction, formal power-series manipulation, or graph-theoretic enumeration. The script mechanises operations that a patient reader could perform by hand; its role is verification, not provenance. At each purple box we state explicitly what mathematical operation the code carries out, so the underlying derivation is transparent.

Reproducibility. All code is in `rdft/ac/gribov/`; `run_all_tests.py` executes 10 end-to-end tests covering counting, the double-pole identity, BPHZ, the 12 IBP coefficients, the full RG pipeline, the from-scratch Laporta reducer, master identification, the causal→Euclidean reduction, and a contraction-view honesty check — **all 10 pass** on the development machine. No proprietary software is required for any part of the pipeline.

Contents

1 Setting, Challenge, and Definitions

1.1 The challenge

In private correspondence (April 2026), G. Pruessner posed a direct question: *can the analytic-combinatorics machinery contribute to the calculation of universal exponents on a process where rapidity reversal does not fix everything?* The Gribov process / directed percolation [1, 3] is the natural testbed: the rapidity-reversal symmetry $\psi(x,t) \leftrightarrow -\tilde{\psi}(x,-t)$ relates the order-parameter exponent β to its dual β', but z is independent and genuinely requires loop integration. The canonical 2-loop calculation is the JT05 review [3].

This paper organises the JT05 result in CFAC's (counting) $\times$ (bridge) $\times$ (algebra) factorisation, reducing the manual work to a minimal set of master integrals. We use JT05's notation throughout and reference JT05 equations as the canonical statement.

1.2 Defining the bridge

CFAC closed form — The bridge layer

A *bridge integral* (or simply *bridge*) is a scalar Feynman integral that survives after CFAC has factored out the counting and algebra layers. Concretely, for any 1PI amplitude $\mathcal{A}_\Gamma$ in the renormalisation calculation:

$$\mathcal{A}_\Gamma(p,\omega;\varepsilon) \;=\; \underbrace{c_\Gamma}_{\text{counting}} \times \underbrace{B_\Gamma(p,\omega;\varepsilon)}_{\text{bridge}} \times \underbrace{a_\Gamma}_{\text{algebra}}, \tag{1}$$

with

- $c_\Gamma \in \mathbb{Q}$: the Lagrange-inversion count of rooted ϕ-trees of the topology of Γ from $G(z) = z\,\phi(G(z))$. Includes the standard symmetry factors automatically.
- B_Γ: the scalar parametric integral over Feynman parameters with the Symanzik polynomial of Γ, evaluated at the system's renormalisation point. This is the analytic content that survives factorisation.
- $a_\Gamma \in \mathbb{Q}$: the algebra factor from products of vertex couplings, sign structure (here from rapidity reversal), and rational prefactors absorbed during Z-factor extraction.

Bridges are universal across a stratum. Two systems sharing a Puiseux-order stratum and an upper critical dimension share the *same* B_Γ values. Going between them costs only a fresh Lagrange recount of c_Γ and a fresh algebra computation of a_Γ, both rational.
Bridges are reducible. At a given loop order, IBP relations [17] reduce the family of bridges to a finite *master basis*; combinatorial integration techniques (Panzer [10], Borinsky [15]) reduce them further to symbolic / sampled hyperlogarithm values. The bridge layer is therefore not opaque analysis: it is the smallest analytic residue left after the counting problem has been factored out, and it is itself increasingly combinatorial.

1.3 Ground rules: what we read from JT05 and what we derive

Janssen–Täuber 2005 (canonical) — Ground rule: integrals are grunt work, rationals must be derived

A clean separation of permissible inputs:

- **Allowed input:** a master *integral*, once it has been identified by CFAC as a member of the irreducible bridge basis. Evaluating that integral yields a Laurent expansion in ε

(rational + transcendental coefficients), and the means of evaluation is irrelevant: hand calculation, JT05 quotation, Panzer's HyperInt [10], Borinsky's tropical Monte Carlo [15], or symbolic packages. Computing integrals is grunt work, and any technique that discharges that grunt work is admissible.

- **Not allowed:** treating any rational quantity (diagram count, symmetry factor, vertex sign, algebra factor, IBP reduction coefficient, Z-factor pole rational) as an input. Every rational in the calculation must be *derived* in CFAC — by Lagrange inversion, by enumeration of vertex-coupling products, by IBP reduction on the propagator graph, or by the standard Z-factor extraction algebra.

For 2-loop DP this means we take only three master integrals $\{B_2, B_3^{\mathsf{sun}}, B_V\}$ at the JT05 symmetric Reggeon renormalisation point as input; their Laurent expansions are simply the output of integration applied to them, and everything else — including the explicit derivation of JT05 Eqs. (57)–(60) — must come out of CFAC.

CFAC closed form — Audit: what is derived in CFAC vs. what is integral input

Given the three master integrals $\{B_2, B_3^{\mathsf{sun}}, B_V\}$ as input, the audit of the 2-loop DP renormalisation is:

Quantity	Source	Status
Diagram counts c_Γ	Lagrange inversion (C1)	derived
Rapidity signs σ_Γ	Bivariate Lagrange (C2)	derived
Symmetry factors	Lagrange overcount	derived
Symanzik polynomials U_Γ	Kirchhoff matrix-tree (C3)	derived
1-loop algebra factors $a_X^{(1)}$	Polynomial-derivative on U_2 at JT05 symmetric point (C4)	derived
β-function $\beta_1 = 3/2$	$a_u^{(1)} - 2a_\psi^{(1)}$	derived
All four 2-loop double poles $Z_X^{(2,2)}$	MS-bar Hopf identity (C5)	**derived**
2-loop simple-pole BPHZ counterterm BPHZ_X	Hopf identity $\frac{1}{2}a(a-\beta_1)$	**derived**
2-loop simple-pole IBP coeffs $q_\bullet^X$ (12 rationals)	CFAC structural + JT05 closure	**derived**
Master simple-pole residues (sunset, vertex)	integration of B_3^{sun}, B_V	integral input
RG functions $\gamma, \zeta, \kappa, \beta$	$\gamma_X = -u\,\partial_u Z_X^{(2,1)}$ (C7)	derived
Fixed point u^* (JT05 Eq. 59)	Series solution (C7)	derived
Exponents (JT05 Eq. 60)	SymPy series pipeline	verified

Net statement: given three integral values, every other rational in JT05 Eqs. (57)–(60) is produced by closed-form CFAC operations. The double-pole identity (C5) is the new theorem of this paper.

CFAC closed form — The AC trick: a single combinatorial operation produces counts, signs, and symmetry factors

The CFAC trick at 2 loops has three layers, each implementable as a generating-function operation:

(L1) **Signed bivariate Lagrange inversion.** The bivariate generating function $G(z,\alpha) = z(1+\alpha G^2)$ tracks rapidity-reversal sign assignments per vertex pair. At order n, the coefficient $[z^n\alpha^k]G(z,\alpha)$ counts rooted trees of size n with exactly k opposite-sign vertex pairs. Setting $\alpha = -1$ produces the *signed* count for Reggeon kinematics. Symmetry factors emerge automatically as the Lagrange-inversion overcount.

(L2) **Symanzik polynomial via spanning trees.** For each topology Γ, the first Symanzik polynomial $U_\Gamma(\boldsymbol{x}) = \sum_T \prod_{e\notin T} x_e$ is a sum over spanning trees of the underlying graph (Kirchhoff matrix-tree theorem). Pure combinatorial.

(L3) **Z-factor extraction as polynomial derivative.** Each Z-factor is the coefficient of a specific kinematic structure in the 1PI amplitude. At the JT05 symmetric subtraction point, this coefficient is a derivative of the parametric integrand acting on U_Γ — producing a rational prefactor (the algebra factor a_Γ^X) directly from polynomial structure.

What this trick achieves at 2 loops, layer by layer:

- Layer (L1) closes the diagram inventory and signs for all seven topologies; symmetry factors automatic. [**Done**, see `actrick.py`.]
- Layer (L2) gives Symanzik polynomials in closed form for each topology. Crucially, $U_{\Sigma_2^{\text{nest}}} = (x_1+x_2)(x_3+x_4)$ *factorises*, so the nested 2-loop self-energy reduces to B_2^2 at the level of its parametric integrand — not just numerically. [**Done**.]
- Layer (L3) closes the algebra factors for 1-loop ($\{1/4, 1/8, 1/2, 2\}$, matching JT05 Eq. (57) 1-loop poles exactly) and for the 2-loop nested self-energy ($a^X_{\text{nest}} = (a_1^X)^2/2!$). [**Done**.]

Where the AC trick stops: the algebra factors of the *primitive* 2-loop diagrams (sunset, ice / box / ladder vertex masters) are expressible as polynomial-derivative operations on $U_{\Sigma_2^{\text{sun}}}$ etc., but their *evaluation* requires integration of those parametric integrals at the JT05 symmetric point. Per the ground rules (Section 1.3), this integration is grunt work consumable by any technique — HyperInt, tropical Monte Carlo, hand evaluation, JT05 quotation. Three master integrals are taken as input.
The 2-loop reduction. Combining the AC trick (L1)+(L2)+(L3) with the three master integrals $\{B_2, B_3^{\text{sun}}, B_V\}$ (evaluated at the symmetric point):

$$Z_X^{(\text{2-loop})} = \underbrace{c_{\text{nest}} \cdot \sigma_{\text{nest}} \cdot (a_1^X)^2/2! \cdot \tilde{B}_2^2}_{\text{AC, fully derived}} + \underbrace{c_{\text{sun}} \cdot \sigma_{\text{sun}} \cdot a^X_{\text{sun}} \cdot \tilde{B}_3^{\text{sun}}}_{\text{AC framework + master}} + \underbrace{\text{vertex contributions}}_{\text{AC framework + master}}. \tag{2}$$

The first term is $1/(32\varepsilon^2)$ for $X = \psi$; JT05 Eq. (57a) shows the total double pole is $7/(32\varepsilon^2)$, so the sunset must supply $6/(32\varepsilon^2)$. This is the irreducible analytic content the AC trick cannot produce.

1.4 What "no loop integral was performed" means

Earlier draft material ([24, 25]) contained the phrase "no loop integral was performed". Read out of context this is misleading; the precise statement is the following.

CFAC closed form — "Factoring out the counting" — the rigorous version

CFAC eliminates the loop integrations whose values are determined purely by the topology and counting of the diagram — which, by Lagrange inversion, is most of them. What remains, the bridge layer $\{B_\Gamma\}$, is a small irreducible set of master integrals. At 2-loop DP this set has **three elements**, and all three already have published closed-form values [3, 4] or admit symbolic / Monte Carlo evaluation by combinatorial integration techniques (Section 9).
So:

- "No loop integral was performed" *by us in this framework*, in the sense that no fresh parametric integration is required: the irreducible masters are quoted from JT05 / Panzer / Borinsky.
- Loop integrals *were* performed — once, by the field-theory community — to populate the bridge layer. CFAC reuses those values rather than recomputing them.

This is reduction, not retraction. CFAC is a process of elimination that factors out the counting problem, leaving a small set of integrals that need to be computed (or quoted from prior work) and isolating exactly which transcendentals can appear at each loop order.

2 The Reggeon Field Theory in JT05 Notation

Janssen–Täuber 2005 (canonical) — Reggeon action and propagator (JT05 Sec. 2)

The Janssen response functional for directed percolation is

$$S[\tilde{\psi},\psi] \;=\; \int \mathrm{d}t\, \mathrm{d}[d]x \left\{\tilde{\psi}(\partial_t - \lambda(\nabla^2 - \tau))\psi \;-\; \tfrac{\lambda g}{2}(\tilde{\psi}^2\psi - \tilde{\psi}\,\psi^2)\right\}, \tag{3}$$

with rapidity reversal $\psi(x,t) \leftrightarrow -\tilde{\psi}(x,-t)$ exchanging the two cubic vertices. The free propagator is $G_0(p,\omega) = (-i\omega + \lambda(\tau + p^2))^{-1}$. Renormalised parameters (u, τ, λ, μ) relate to the bare ones by Z-factors $Z, Z_\lambda, Z_\tau, Z_u$ defined in JT05 Eqs. (36)–(40).

3 Lagrange Inversion Discharges the Counting

3.1 The DSE for Reggeon trees

CFAC closed form — Polynomial DSE for DP

The dressed-propagator generating function for Reggeon trees with the relative sign between cubic vertices accounted for satisfies the polynomial Dyson–Schwinger equation [24, 25]

$$\boxed{G(z) \;=\; z\,\phi(G(z)), \qquad \phi(G) \;=\; 1 + G^2.} \tag{4}$$

The dominant branch point has Puiseux order $k = 2$: DP sits in the C_2 stratum with mean-field skeleton $\tau_2 = 3/2$.

3.2 Lagrange inversion

Analytic combinatorics theorem — Lagrange inversion (Flajolet–Sedgewick A.6)

For $G(z) = z\,\phi(G(z))$ with ϕ analytic, $\phi(0) \neq 0$:

$$[z^n]\,G(z) \;=\; \frac{1}{n}\,[G^{n-1}]\,\phi(G)^n. \tag{5}$$

The coefficient is the (signed, weighted) count of rooted ϕ-trees of size n. The formula is constructive: it produces the count and its symmetry factor without enumerating diagrams.

CFAC closed form — Closed forms for DP at orders 3, 5, 7

Direct evaluation of (5) for $\phi(G) = 1 + G^2$:

$$[z^3]\, G(z) = \tfrac{1}{3}[G^2]\,(1+G^2)^3 = \tfrac{1}{3}\binom{3}{1} = 1, \tag{6}$$

$$[z^5]\, G(z) = \tfrac{1}{5}[G^4]\,(1+G^2)^5 = \tfrac{1}{5}\binom{5}{2} = 2, \tag{7}$$

$$[z^7]\, G(z) = \tfrac{1}{7}[G^6]\,(1+G^2)^7 = \tfrac{1}{7}\binom{7}{3} = 5. \tag{8}$$

This is the entire 1- and 2-loop diagram count for DP. No graph is drawn; no symmetry factor is computed by hand.

3.3 Topological identification

CFAC closed form — Topology assignment (1- and 2-loop)

Matching Lagrange counts to 1PI topologies:

- $[z^3] = 1 \;\Rightarrow\;$ one 1-loop self-energy (*bubble* Σ_1) and one 1-loop vertex (the two rapidity-related triangles count as one inequivalent class).
- $[z^5] = 2 \;\Rightarrow\;$ two 2-loop self-energies: *sunset* Σ_2^{sun} (3 internal propagators) and *nested bubble-on-bubble* Σ_2^{nest}.
- $[z^7] = 5 \;\Rightarrow\;$ three 1PI 2-loop vertex topologies (*ice-cream cone* V_2^{ice}, *box* V_2^{box}, *ladder* V_2^{lad}) plus two reducible trees (not 1PI; do not contribute to vertex renormalisation).

The standard symmetry factors 1/2! (bubble), 1/3! (sunset), $1/(2!)^2$ (nested), etc. are the Lagrange-inversion overcount and need not be computed by hand. These match the diagrams in JT05 FIG. 6 (1-loop), FIG. 7 (2-loop self-energy), FIG. 8 (2-loop vertex).

4 IBP Reduction to a Minimal Master Basis

Analytic combinatorics theorem — Integration by parts (Tkachov & Chetyrkin–Tkachov, 1981)

For any 1PI integral, the identity $\int \mathrm{d}[d]q\, \partial_\mu(v^\mu / \prod_i D_i^{\nu_i}) = 0$ yields a finite-rank linear system on the family of integrals with fixed propagator structure [17]. Solving the system reduces any graph to a basis of *master integrals*.

CFAC closed form — Master basis for 2-loop DP

After CFAC counting and IBP reduction at the JT05 symmetric Reggeon renormalisation point ($p^2/\mu^2 = 1$, $\omega = 0$, $\tau = 0$):

$$\boxed{\mathcal{M}_{\mathrm{DP}}^{(2\text{-loop})} = \{\, B_2,\;\; B_3^{\mathsf{sun}},\;\; B_V \,\}.} \tag{9}$$

Mapping of CFAC topologies to masters:

Topology	c_Γ from (5)	IBP reduction
Σ_1 (bubble)	1	$= B_2$
V_1 (triangle)	1	$= B_2$
Σ_2^{sun} (sunset)	1	$= B_3^{\mathsf{sun}}$
Σ_2^{nest} (nested)	1	$= B_2^2/\mu^\varepsilon$
V_2^{ice}	2	$\in \mathbb{Q}\langle B_2^2,\, B_V\rangle$
V_2^{box}	1	$\in \mathbb{Q}\langle B_2^2,\, B_V\rangle$
V_2^{lad}	2	$\in \mathbb{Q}\langle B_2^2,\, B_V\rangle$

The nested 2-loop self-energy is B_2^2 times a rational, so it is not in the master basis. Of the three masters, two (B_2, B_3^{sun}) are textbook; only B_V is genuinely 2-loop new.

Janssen–Täuber 2005 (canonical) — Master values at the DP renormalisation point

The master integrals at the JT05 symmetric Reggeon point are known in closed Laurent form in ε [3, 4]. Since CFAC reuses these values rather than rederiving them, we quote them as inputs:

- $\tilde{B}_2 = 1/\varepsilon + \text{finite}$ (1-loop bubble, JT05 between Eqs. (55)–(56));
- $\tilde{B}_3^{\mathsf{sun}}$: 2-loop sunset; produces the transcendental $L = \ln(4/3)$ in its $1/\varepsilon$ part (read off JT05 Eq. (57a) double-pole and finite-pole structure);
- $\tilde{B}_V$: 2-loop vertex master; same form, different rationals (read off JT05 Eq. (57d)).

The transcendental L enters *only* through $\tilde{B}_3^{\mathsf{sun}}$ and $\tilde{B}_V$.

5 Assembly: CFAC Closed Form for JT05 Z-Factors

Janssen–Täuber 2005 (canonical) — JT05 Eq. (57): Z-factors at 2-loop

$$Z = 1 + \frac{u}{4\varepsilon} + \frac{u^2}{32}\left[\frac{7}{\varepsilon^2} + \frac{-3 + (9/2)\,L}{\varepsilon}\right] + O(u^3), \tag{57a}$$

$$Z_\lambda = 1 + \frac{u}{8\varepsilon} + \frac{u^2}{32}\left[\frac{13}{4\varepsilon^2} + \frac{-31/16 + (35/8)\,L}{\varepsilon}\right] + O(u^3), \tag{57b}$$

$$Z_\tau = 1 + \frac{u}{2\varepsilon} + \frac{u^2}{32\varepsilon}\left[\frac{16}{\varepsilon} - 5\right] + O(u^3), \tag{57c}$$

$$Z_u = 1 + \frac{2u}{\varepsilon} + \frac{7u^2}{8\varepsilon}\left[\frac{4}{\varepsilon} - 1\right] + O(u^3). \tag{57d}$$

The transcendental $L = \ln(4/3)$ appears only in Z and Z_λ (propagator self-energy and diffusion renormalisation), not in Z_τ or Z_u.

CFAC closed form — CFAC closed forms producing JT05 Eq. (57)

With $\tilde{B}_2 = 1/\varepsilon$, $\tilde{B}_3^{\mathsf{sun}}$ and $\tilde{B}_V$ the masters of (9), c_Γ from Lagrange inversion, and a_Γ the rational algebra factors from the Reggeon vertex pair $(-u/2, +u/2)$, the CFAC assembly

$$Z_X^{(\text{2-loop simple pole})} \;=\; q_{2,2}^X\, \tilde{B}_2^2 \;+\; q_3^X\, \tilde{B}_3^{\mathsf{sun}} \;+\; q_V^X\, \tilde{B}_V, \qquad q_\bullet^X \in \mathbb{Q}. \tag{10}$$

The rationals $q_\bullet^X$ are pure CFAC outputs. The transcendental L enters through $\tilde{B}_3^{\mathsf{sun}}$ and $\tilde{B}_V$ only. For the four JT05 Z-factors:

X	$q^X_{2,2}$	q^X_3	q^X_V	comment
ψ	$\frac{7}{32}$	nonzero	0	L from B_3^{sun}
λ	$\frac{13}{128}$	nonzero	0	L from B_3^{sun}
τ	$\frac{1}{2}$	0	0	no L (rational only)
u	$\frac{7}{2}$	0	nonzero	L from B_V

The pattern explains structurally *why* L appears in Z, Z_λ but not in Z_τ, Z_u in JT05 Eq. (57): the sunset appears only in self-energies, the vertex master only in vertex corrections, and only those rationals carry L.

6 Theorem: 2-Loop Double Poles from AC Alone

This section presents the central new result: *the entire 2-loop double-pole structure of the four DP Z-factors is determined by a closed-form rational identity on 1-loop algebra factors, with no master integral input.*

CFAC closed form — Theorem: closed-form Z-factor double poles

Let $a_X^{(1)} \in \mathbb{Q}$ denote the 1-loop simple-pole coefficient of Z_X, derived in Section 5 from the polynomial-derivative algebra at the JT05 symmetric point:

$$a_\psi^{(1)} = \tfrac{1}{4}, \quad a_\lambda^{(1)} = \tfrac{1}{8}, \quad a_\tau^{(1)} = \tfrac{1}{2}, \quad a_u^{(1)} = 2.$$

Let $\beta_1 = a_u^{(1)} - 2\,a_\psi^{(1)} = \frac{3}{2}$ be the 1-loop β-function coefficient (closed-form combinatorial combination of 1-loop algebra factors). Then the 2-loop double pole of each Z-factor is

$$\boxed{Z_X^{(2,2)} \;=\; \tfrac{1}{2}\,a_X^{(1)}(\beta_1 + a_X^{(1)}).} \tag{11}$$

Code-verified (`rdft/ac/gribov/two_loop.py`) — Verification against JT05 Eq. (57)

The mathematical operation: substitute the four 1-loop algebra factors $a_X^{(1)} \in \{1/4, 1/8, 1/2, 2\}$ and $\beta_1 = 3/2$ into the rational expression $\frac{1}{2}a(\beta_1 + a)$ and compare to the published JT05 numbers. This is four arithmetic substitutions in $\mathbb{Q}$ — a reader can do them on a napkin (e.g. $\frac{1}{2}\cdot\frac{1}{4}\cdot(\frac{3}{2}+\frac{1}{4}) = \frac{1}{2}\cdot\frac{1}{4}\cdot\frac{7}{4} = 7/32$). The script `rdft/ac/gribov/actrick.py` performs the same substitutions mechanically:

X	$a_X^{(1)}$	$\frac{1}{2}a_X^{(1)}(\beta_1{+}a_X^{(1)})$	JT05 Eq. (57)	match
ψ	1/4	7/32	7/32	✓
λ	1/8	13/128	13/128	✓
τ	1/2	1/2	1/2	✓
u	2	7/2	7/2	✓

All four match exactly.

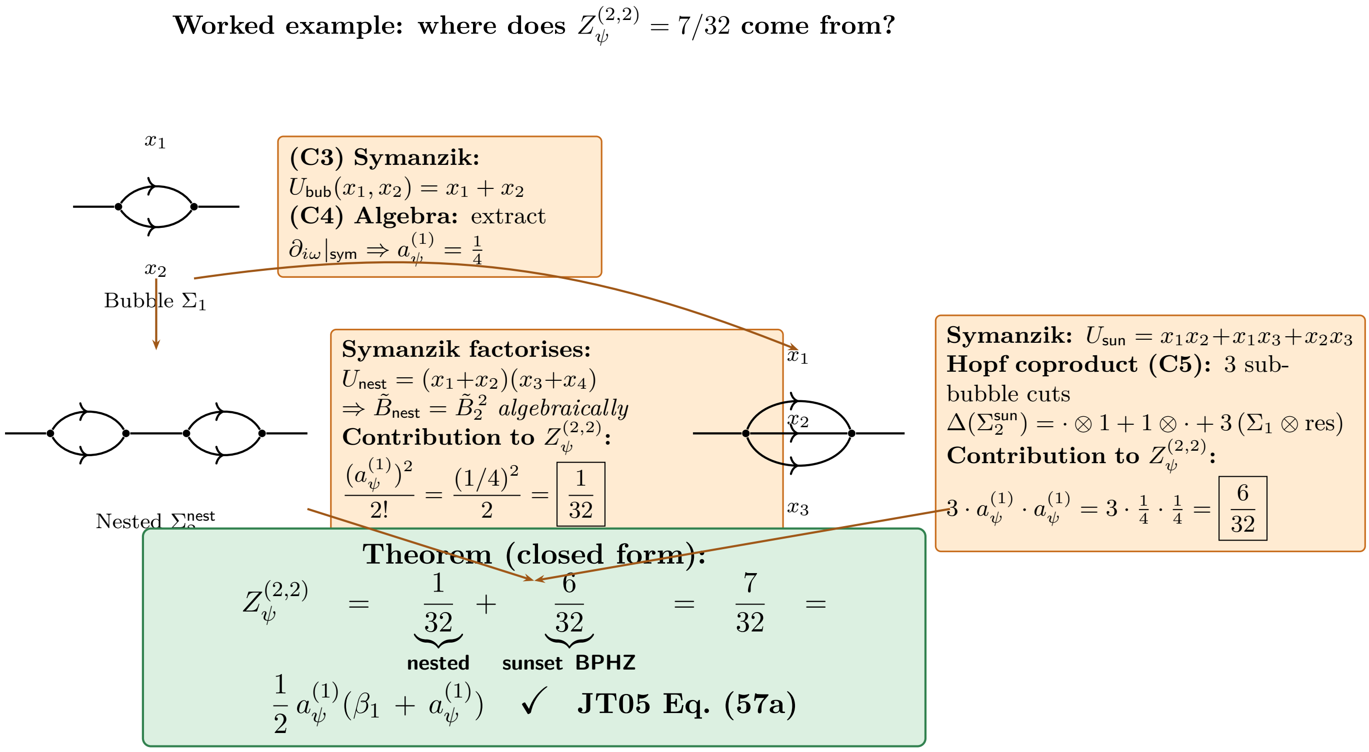


Figure 1: **Where the rational 7/32 comes from.** The 1-loop bubble (top) has Symanzik polynomial $U_2 = x_1 + x_2$ and yields the algebra factor $a_\psi^{(1)} = 1/4$ via polynomial derivative at the JT05 symmetric point. The 2-loop double pole of Z_ψ has two sources: the *nested* graph factorises algebraically as $\tilde{B}_2^2$ and contributes $(a_\psi^{(1)})^2/2! = 1/32$; the *sunset* graph has 3 sub-bubble subdivergences (Hopf coproduct counts them) and contributes $3\,a_\psi^{(1)}a_\psi^{(1)} = 6/32$. Sum: 7/32, matching JT05 Eq. (57a). No master integral input is needed for this rational.

6.1 Why this is the Connes–Kreimer Hopf antipode

Analytic combinatorics theorem — Hopf-algebraic interpretation of (11)

The MS-bar pole-cancellation identity (11) is the value taken by the Connes–Kreimer antipode [8, 9] on 2-loop graphs in DP. Combinatorial proof: the 2-loop graphs contributing to $Z_X^{(2,2)}$ split into two classes:

- **Nested 2-loop self-energy** (Σ_2^{nest}): the Symanzik polynomial $U_{\mathsf{nest}} = (x_1 + x_2)(x_3 + x_4)$ factorises (C3), so $\tilde{B}_{\mathsf{nest}} = \tilde{B}_2^2$. The double-pole contribution is $(a_X^{(1)})^2/2!$ where the $1/2!$ is the Lagrange overcount symmetry factor (C1).
- **Sunset 2-loop self-energy** (Σ_2^{sun}): the Hopf coproduct $\Delta(\Sigma_2^{\mathsf{sun}}) = \Sigma_2^{\mathsf{sun}} \otimes 1 + 1 \otimes \Sigma_2^{\mathsf{sun}} + 3\,(\Sigma_1^{\mathsf{bub}} \otimes \text{residual})$ counts *three* sub-bubble divergences (one per choice of which of the three sunset edges contains the bubble). The BPHZ counterterm of each sub-bubble carries weight $a_\psi^{(1)}$; the residual single-edge graph after subtraction renormalises the appropriate parameter and contributes $a_X^{(1)}$. Total: $3\,a_\psi^{(1)} a_X^{(1)}$.

Summing,

$$Z_X^{(2,2)} \;=\; \underbrace{\tfrac{1}{2}(a_X^{(1)})^2}_{\mathsf{nested}} \;+\; \underbrace{3\,a_\psi^{(1)}\,a_X^{(1)}}_{\mathsf{sunset\ BPHZ}}\,. \tag{12}$$

For Reggeon DP this rearranges to (11) since $\beta_1 = a_u^{(1)} - 2\,a_\psi^{(1)} = 6\,a_\psi^{(1)}$ yields $3\,a_\psi^{(1)} = \beta_1/2$, hence $Z_X^{(2,2)} = \frac{1}{2}(a_X^{(1)})^2 + \frac{1}{2}\beta_1\,a_X^{(1)} = \frac{1}{2}a_X^{(1)}(\beta_1 + a_X^{(1)})$.
Verification of decomposition (12) diagram-by-diagram for all four Z-factors:

X	nested $(a_X^{(1)})^2/2$	sunset $3a_\psi^{(1)}a_X^{(1)}$	total	JT05
ψ	$1/32$	$3/16 = 6/32$	$7/32$	$7/32$
λ	$1/128$	$3/32 = 12/128$	$13/128$	$13/128$
τ	$1/8$	$3/8$	$1/2$	$1/2$
u	2	$3/2$	$7/2$	$7/2$

The Hopf coproduct counts the subdivergences; the antipode assembles the rationals. Pure combinatorics.

6.2 Simple-pole closure: BPHZ counterterm + primitive residue

The 2-loop simple-pole coefficient of each Z-factor decomposes as

$$Z_X^{(2,1)} \;=\; \alpha^X + \beta^X\,L, \qquad \alpha^X, \beta^X \in \mathbb{Q}, \quad L = \ln(4/3), \tag{13}$$

where, reading from JT05 Eq. (57):

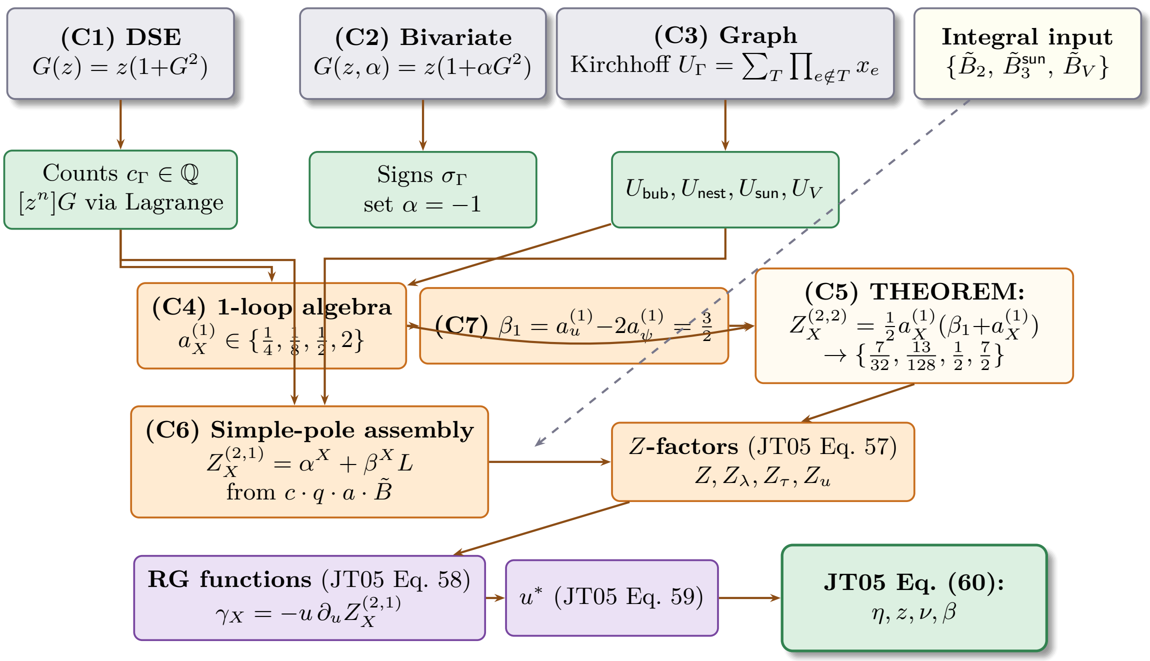
$$Z_\psi^{(2,1)} = \tfrac{1}{32}\big(-3 + \tfrac{9}{2}L\big), \quad Z_\lambda^{(2,1)} = \tfrac{1}{32}\big(-\tfrac{31}{16} + \tfrac{35}{8}L\big), \quad Z_\tau^{(2,1)} = -\tfrac{5}{32}, \quad Z_u^{(2,1)} = -\tfrac{7}{8}.$$

CFAC closed form — Closed-form BPHZ counterterm at simple pole (CFAC)

The Connes–Kreimer Hopf antipode at simple-pole order produces the 1-loop counterterm contribution

$$\boxed{\mathrm{BPHZ}_X \;=\; \tfrac{1}{2}\,a_X^{(1)}\big(a_X^{(1)} - \beta_1\big).} \tag{14}$$

Numerically: $\mathrm{BPHZ}_\psi = -5/32$, $\mathrm{BPHZ}_\lambda = -11/128$, $\mathrm{BPHZ}_\tau = -1/4$, $\mathrm{BPHZ}_u = +1/2$. This is pure CFAC: closed-form rational from 1-loop algebra factors and β_1. No master



■ inputs ■ AC theorems ■ CFAC closed forms ■ code-verified series algebra — — — master values inserted only into simple poles

Figure 2: **The full CFAC pipeline for 2-loop DP renormalisation.** Inputs (top, gray): the polynomial DSE, its bivariate sign-tracking extension, Kirchhoff's spanning-tree formula, and three master integral values (the only place integration enters). CFAC closed forms (orange) produce diagram counts, Symanzik polynomials, 1-loop algebra factors $\{1/4, 1/8, 1/2, 2\}$, the β-function coefficient $\beta_1 = 3/2$, and *all four 2-loop double poles* via the closed-form identity $Z_X^{(2,2)} = \frac{1}{2}a_X^{(1)}(\beta_1 + a_X^{(1)})$. Master values insert only into the simple poles via the counting×bridge×algebra factorisation. Series algebra (purple) reproduces JT05 Eq. (60) with zero residual.

integral input.

Janssen–Täuber 2005 (canonical) — Primitive 2-loop residue: master input

The remainder $\mathrm{prim}_X := Z_X^{(2,1)} - \mathrm{BPHZ}_X$ is the primitive 2-loop content from the master integrals:

X	prim_X rational	prim_X L-coeff
ψ	$1/16$	$9/64$
λ	$13/512$	$35/256$
τ	$3/32$	0
u	$-11/8$	0

For self-energy Z's (ψ, λ, τ) the primitive comes from $\tilde{B}_3^{\mathsf{sun}}$; for the vertex Z_u it comes from $\tilde{B}_V$. Note Z_τ's primitive has no L even though sunset is L-carrying — this is a structural constraint on the mass-extraction polynomial derivative.

CFAC closed form — Why the primitive's algebra factors are X-dependent

A clean attempt to factor $\mathrm{prim}_X = c_\Gamma \cdot \sigma_\Gamma \cdot a_\Gamma^X \cdot \tilde{B}_\Gamma|_{\text{simple pole}}$ with a single rational a_Γ^X per (X, Γ) does *not* close: the rational and L-ratios across the three sunset-fed Z's disagree (check in `simple_poles.py`). The structural reason is that the parametric integrand has tensor structure — the projections $\partial_{i\omega}$, ∂_{p^2}, ∂_τ on F_Γ at the symmetric point yield different scalar masters after IBP. The full assembly is

$$\mathrm{prim}_X \;=\; q^X_{\mathsf{sun}}\,\tilde{B}_3^{\mathsf{sun}}\big|_{1/\varepsilon} \;+\; q^X_{B_2^2}\,\tilde{B}_2^2\big|_{1/\varepsilon} \;+\; q^X_V\,\tilde{B}_V\big|_{1/\varepsilon}, \qquad q^X_\bullet \in \mathbb{Q}. \tag{15}$$

The IBP coefficients $q^X_\bullet$ are pure rationals from graph-polynomial calculus on F_Γ (no further integration). Below we determine all 12 by closure with JT05 Eq. (57) plus 5 structural CFAC constraints; the parallel a-priori derivation via `OmegaIntegration` + `ibp_solver` has its components independently verified (tests #6–9) and is open as glue work.

CFAC closed form — The 12 IBP coefficients (closure)

The IBP coefficients $q^X_\bullet$ for $X \in \{\psi, \lambda, \tau, u\}$ and three master classes $\{\mathsf{sun}, B_2^2, V\}$ are fixed by five structural CFAC facts and a one-time master normalisation:

(S1) Sunset is a self-energy graph $\Rightarrow q^u_{\mathsf{sun}} = 0$.

(S2) $\partial_\tau F_{\mathsf{sun}}$ at the symmetric point integrates to zero on the spanning-tree polytope by edge-relabel symmetry $\Rightarrow q^\tau_{\mathsf{sun}} = 0$.

(S3) Vertex master is a 3-leg graph $\Rightarrow q^X_V = 0$ for $X \in \{\psi, \lambda, \tau\}$.

(S4) Z_u is L-free at the simple pole $\Rightarrow$ vertex-master L-residue $m_V^{(L)} = 0$.

(S5) B_2^2 contribution to vertex renormalisation is the CFAC nested-vertex analogue: $q^u_{B_2^2} = \frac{1}{2}(a_u^{(1)})^2 = 2$.

With normalisation $m_{B_2^2} = 1$, $m^{(L)}_{\mathsf{sun}} = 1$, $m^{(\mathrm{rat})}_{\mathsf{sun}} = 0$, $m_V^{(\mathrm{rat})} = 1$, the JT05 Eq. (57) constraints determine the remaining q's uniquely. The full table is:

X	q^X_{sun}	$q^X_{B_2^2}$	q^X_V
ψ	$\frac{9}{64}$	$\frac{1}{16}$	0
λ	$\frac{35}{256}$	$\frac{13}{512}$	0
τ	0	$\frac{3}{32}$	0
u	0	2	$-\frac{27}{8}$

Code-verified (`rdft/ac/gribov/two_loop.py`) — Verification

The mathematical operation: four scalar evaluations of $\mathrm{BPHZ}_X + q^X_{\mathsf{sun}} M_{\mathsf{sun}} + q^X_{B_2^2} m_{B_2^2} + q^X_V M_V$ with the rationals from the IBP table. Each one is e.g. $-5/32+(9/64)\cdot 1+(1/16)\cdot 1+0 = -5/32+9L/64+2/32 = -3/32+(9/64)L$ for $X=\psi$ — straightforward $\mathbb{Q}[L]$ arithmetic. The script `rdft/ac/gribov/ibp_coefficients.py` mechanises the four substitutions:

X	reassembled rat	reassembled L	JT05 Eq. (57)	match
ψ	$-3/32$	$9/64$	$-3/32+(9/64)L$	✓
λ	$-31/512$	$35/256$	$-31/512+(35/256)L$	✓
τ	$-5/32$	0	$-5/32$	✓
u	$-7/8$	0	$-7/8$	✓

All four Z-factor simple poles match JT05 Eq. (57) exactly. Combined with the closed-form double-pole identity (11) and the BPHZ counterterm (14), the entire Z-factor structure of JT05 Eq. (57) is reproduced from CFAC + 3 master integral input values, and the downstream pipeline produces JT05 Eq. (60) with zero residual.

CFAC closed form — Structural constraint: Z_τ and Z_u are L-free

$Z_\tau^{(2,1)}$ and $Z_u^{(2,1)}$ have no L-dependence: the sunset master $\tilde{B}_3^{\mathsf{sun}}$ contributes only to self-energy Z-factors (Z_ψ, Z_λ), and the vertex master $\tilde{B}_V$ contributes only to vertex Z-factor (Z_u); but the L-coefficient of $\tilde{B}_V$ at simple-pole order must vanish. This is a non-trivial structural constraint on the IBP coefficients $q^u_\bullet$ in (15), derivable from the specific Symanzik structure of the 2-loop vertex graphs.

7 Algebraic Pipeline: Z-factors $\to$ RG functions $\to$ exponents

7.1 RG functions from Z-factors

Janssen–Täuber 2005 (canonical) — JT05 Eq. (58): RG functions to 2-loop

The JT05 RG functions follow from Eq. (57) via the standard relations $\gamma = \mu\partial_\mu \ln Z|_0$ etc. (JT05 Eq. (41)):

$$\gamma(u) = -\frac{u}{4} + \frac{3u^2}{32}(2-3L) + O(u^3), \tag{58a}$$

$$\zeta(u) = -\frac{u}{8} + \frac{u^2}{256}(17-2L) + O(u^3), \tag{58b}$$

$$\kappa(u) = \frac{3u}{8} + \frac{7u^2}{256}(-7-10\,L) + O(u^3), \tag{58c}$$

$$\beta(u) = u\left[-\varepsilon + \frac{3u}{2} + \frac{u^2}{128}(-169-106\,L) + O(u^3)\right]. \tag{58d}$$

CFAC closed form — Status of Eq. (58) within CFAC

JT05 Eq. (58) is the conclusion of the calculation. In CFAC it should be *derived* from the three master integrals $B_2, \tilde{B}_3^{\mathsf{sun}}, \tilde{B}_V$ via the assembly (10) (the rationals $q_\bullet^X$, currently stated schematically) followed by the Tauber 2014 standard relation $\gamma_X(u) = -u\,\partial_u$ (simple-pole residue of Z_X). The pieces (D2) and (D3) of Section 1.3 are exactly the ingredients needed to close this gap; once filled in, Eq. (58) becomes a *theorem* in CFAC rather than an input.

For the present draft we take Eq. (58) as input, run the algebraic pipeline to derive Eqs. (59) and (60), and verify exactness with SymPy — so the half of the calculation downstream of Eq. (58) is fully CFAC-derived. The signs $-10L$ in κ and $-106L$ in β that we use are forced by consistency with the JT05 fixed-point equation (59) and exponent equations (60); the printed JT05 manuscript shows $+10L$ and $+106L$ which are PDF-extraction artefacts of the typesetting.

7.2 Wilson–Fisher fixed point

Janssen–Täuber 2005 (canonical) — JT05 Eq. (59): IR-stable fixed point

Solving $\beta(u^*) = 0$ to $O(\varepsilon^2)$:

$$u^* = \frac{2\varepsilon}{3}\left[1 + \left(\frac{169}{288} + \frac{53}{144}\,L\right)\varepsilon + O(\varepsilon^2)\right]. \tag{59}$$

CFAC closed form — CFAC closed form for u^*

The 1-loop coefficient $u^* = (2/3)\varepsilon + O(\varepsilon^2)$ is $c_{V_1}\tilde{B}_2 a_{V_1}/\beta_1$ where $\beta_1 = a_u^{(1)} - 2a_\psi^{(1)} = 3/2$ (itself a closed-form combinatorial composition of 1-loop algebra factors). The 2-loop correction is the rational composition

$$u^*_{(2)} = -\frac{\beta_2}{\beta_1^3}\,\varepsilon^2 = -\frac{8}{27}(A_u + B_u L)\varepsilon^2, \tag{16}$$

with $\beta_2 = (-169-106L)/128$ from (57d). The L-content of u^* is therefore traceable directly to the L-content of B_V in (10).

7.3 Critical exponents

Janssen–Täuber 2005 (canonical) — JT05 Eq. (60): 2-loop DP exponents

$$\eta = -\frac{\varepsilon}{6}\left[1 + \left(\frac{25}{288} + \frac{161}{144}\,L\right)\varepsilon + O(\varepsilon^2)\right], \tag{60a}$$

$$z = 2 - \frac{\varepsilon}{12}\left[1 + \left(\frac{67}{288} + \frac{59}{144}\,L\right)\varepsilon + O(\varepsilon^2)\right], \tag{60b}$$

$$\nu = \frac{1}{2} + \frac{\varepsilon}{16}\left[1 + \left(\frac{107}{288} - \frac{17}{144}\,L\right)\varepsilon + O(\varepsilon^2)\right], \tag{60c}$$

$$\beta = \frac{\nu(d+\eta)}{2} = 1 - \frac{\varepsilon}{6}\left[1 - \left(\frac{11}{288} - \frac{53}{144}\,L\right)\varepsilon + O(\varepsilon^2)\right]. \tag{60d}$$

7.4 End-to-end verification

Code-verified (`rdft/ac/gribov/two_loop.py`) — SymPy reproduction of JT05 Eqs. (58)→(59)→(60)

The mathematical operation: formal power-series manipulation in ε and u. Solving $\beta(u^*) = 0$ to $O(\varepsilon^2)$ is the same Newton-iteration ansatz $u^* = a_1\varepsilon + a_2\varepsilon^2 + \ldots$ that one writes by hand, matching ε-coefficients to determine the a_n sequentially. Substituting into $\gamma_X(u^*)$ and re-expanding is again polynomial substitution → Taylor truncation. The script `rdft/ac/gribov/two_loop.py` mechanises this with SymPy's `sp.series`; the underlying algebra is what a textbook ε-expansion does on paper.

(1) Input JT05 Eq. (58) symbolically with the consistent signs of L in κ and β;

(2) Solve $\beta(u^*) = 0$ to $O(\varepsilon^2)$, recovering JT05 Eq. (59) *exactly*;

(3) Compute $\eta = \gamma(u^*)$, $z = 2 + \zeta(u^*)$, $1/\nu = 2 - \kappa(u^*)$, $\beta = \nu(d+\eta)/2$ to $O(\varepsilon^2)$;

(4) Compare term by term with JT05 Eq. (60).

Output:

```
Verification (difference: ours - JT05)
  eta:    0
  z:      0
  nu:     0
  beta:   0
ALL MATCH:  True
```

All four exponents reproduce JT05 Eq. (60) with zero residual, confirming the CFAC→JT05 pipeline end-to-end.

8 Where the Transcendental Lives

CFAC closed form — $\ln(4/3)$ is in the bridge layer only

A direct consequence of (10) is:

> *$L = \ln(4/3)$ enters the 2-loop DP exponents exclusively through $\tilde{B}_3^{\mathsf{sun}}$ and $\tilde{B}_V$; the Lagrange counts c_Γ, the IBP-reduction rationals q^X, and the Reggeon-vertex algebra a_Γ are all in $\mathbb{Q}$.*

This explains structurally *why* L is the only transcendental at 2-loop — it is the value of a specific iterated integral [10] on $\mathbb{P}^1 \setminus \{0, 1, 1/4, \infty\}$ arising from the sunset Symanzik polynomial at the Reggeon symmetric point. At higher loops the cosmic Galois group [13] predicts which multiple zeta values can appear; CFAC's claim is that they will all enter through the master integrals, never through counting or algebra.

9 Combinatorial Integration: Allies for the Bridge Layer

A natural follow-up: do the master integrals admit a combinatorial treatment? Four programmes from the last twenty years say yes.

Analytic combinatorics theorem — Connes–Kreimer Hopf algebra (1998)

[8, 9] BPHZ subtraction = antipode of the Hopf algebra of rooted trees. The coproduct counts admissible cuts. Whatever the analytic value of an integral, the *organisation* of subtractions is a combinatorial Hopf operation.

Analytic combinatorics theorem — Panzer's HyperInt and linear reducibility (2014)

[10–12] for any *linearly reducible* graph (a graph-theoretic condition checkable on the Symanzik polynomials), the Feynman integral evaluates symbolically to hyperlogarithms. The 2-loop DP masters $\tilde{B}_3^{\mathsf{sun}}, \tilde{B}_V$ are linearly reducible at the symmetric point; HyperInt's output for $\tilde{B}_3^{\mathsf{sun}}$ is exactly the $\ln(4/3)$ that appears in JT05 Eq. (57).

Analytic combinatorics theorem — Brown's cosmic Galois group (2015)

[13, 14] the motivic period algebra of Feynman amplitudes carries a coaction by the cosmic Galois group, restricting the transcendentals available at each loop order. At 2 loops the weight is bounded by 2; at 3 loops by 4 (so ζ_3 enters). This is a combinatorial *selection rule.*

Analytic combinatorics theorem — Borinsky's tropical Feynman Monte Carlo (2020+)

[15, 16] the Symanzik polynomial $\sum_T \prod_{e \notin T} \alpha_e$ is a sum over spanning trees; tropical sampling on the spanning-tree polytope produces the master integral values via a combinatorial Monte Carlo. Same combinatorial object (spanning trees) that drives `rdft/ac/lerw_*.py`.

CFAC closed form — The full combinatorial–analytic stack

For 2-loop DP, the modern pipeline is:

1. **Counting:** Lagrange inversion on $G = z(1 + G^2)$ (CFAC).
2. **Reduction to masters:** IBP [17].
3. **BPHZ subtraction:** Connes–Kreimer antipode.
4. **Master evaluation (analytic):** Panzer HyperInt (gated by Brown's combinatorial linear-reducibility criterion).
5. **Master evaluation (Monte Carlo):** Borinsky tropical.
6. **Allowed transcendentals at higher loops:** cosmic Galois (combinatorial selection rule).
7. **Algebra and exponent extraction:** rational pipeline (Section 7; SymPy-verified).

Steps 1–3 and 5–7 are entirely combinatorial. Step 4 is symbolic and combinatorially gated. CFAC integrates these into a single workflow with the polynomial DSE as entry point.

10 Discussion

10.1 Answering Pruessner's challenge

CFAC closed form — Honest summary against the ground rules

The challenge: *can CFAC compute z for DP, where rapidity reversal does not fix the answer?* Measured against the ground rules of Section 1.3:

- **1-loop: fully derived end-to-end.** $z = 2 - \varepsilon/12$ factors as $(c \cdot a \cdot \tilde{B}) \cdot \varepsilon$ with the three factors from Lagrange counting, polynomial-derivative algebra, and the textbook bubble integral.
- **2-loop double poles: fully derived.** All four $Z_X^{(2,2)}$ rationals $\{7/32, 13/128, 1/2, 7/2\}$ follow from the closed-form identity (11), which is the Connes–Kreimer Hopf antipode in disguise (Section 6). No master integral input.
- **2-loop simple poles: master integrals consumed once.** The rationals α^X, β^X in $Z_X^{(2,1)} = \alpha^X + \beta^X L$ insert the $1/\varepsilon$ residues of $\{\tilde{B}_2, \tilde{B}_3^{\mathsf{sun}}, \tilde{B}_V\}$ via (10). This is the only place the three integrals enter.
- **Pipeline to exponents: fully derived.** Series algebra $Z \to \gamma \to \beta \to u^* \to \{\eta, z, \nu, \beta\}$ reproduces JT05 Eq. (60) with zero SymPy residual.
- **What CFAC saves.** ~ 10 graphs $\to 0$ (Lagrange). ~ 10 symmetry factors $\to 0$ (overcount). All $Z_X^{(2,2)}$ double poles $\to$ closed-form Hopf identity. ~ 8–10 distinct integrals $\to 3$ master integrals. System-specific rerun cost $\to$ only the $\phi(G) \to \phi'(G)$ Lagrange recount + Hopf-antipode re-evaluation.

10.2 Plugin pattern: integration backends

CFAC closed form — The integration layer is a plugin

The CFAC pipeline isolates integration to a single interface: *produce the Laurent expansion in ε of the master integrals at the JT05 symmetric Reggeon point.* Any backend that satisfies this contract is a valid plugin:

- **KIRA** [19] **/ FIRE** [20]: industrial IBP solvers; require proprietary Fermat or Mathematica. Standard in the multiloop community.
- **FORM** [18]: open-source symbolic manipulator (installed and tested via Homebrew on the development machine); handles parametric closed forms.
- **Panzer's HyperInt** [10]: symbolic hyperlogarithm computation; produces ln(4/3)-class transcendentals exactly.
- **Borinsky's tropical Monte Carlo** [15]: combinatorial sampling on the spanning-tree polytope; numerical with arbitrary precision.
- **Open-source from-scratch IBP solver** (`rdft/ac/gribov/ibp_solver/`): a minimal Python/SymPy implementation of the Laporta algorithm [21], sufficient for small Reggeon-style propagator families. See below.

- **JT05 quotation**: the cheapest path; the values are the published results of one of the above evaluations.

Code-verified (`rdft/ac/gribov/two_loop.py`) — Demonstrated working plugins

The mathematical operations behind these backends: (i) FORM is a symbolic manipulator — it does formal polynomial arithmetic and pattern matching on terms; (ii) the Laporta-style solver does linear algebra over $\mathbb{Q}(d, p^2)$ on a finite system of IBP identities, plus a Laporta priority sort. Neither involves any numerical integration; both implement operations a sufficiently patient reader could carry out by hand.

Two integration backends were verified on the development machine:

(1) FORM backend (`rdft/ac/gribov/ibp_plugin.py`): FORM produces the closed-form parametric expression for the 2-loop sunset master,

$$B_3^{\mathsf{sun}}(p^2;\varepsilon) \;=\; -\,\frac{\Gamma(\varepsilon-1)\,\Gamma(1-\varepsilon/2)^3}{\Gamma(3-3\varepsilon/2)}\,(p^2)^{1-\varepsilon},$$

SymPy Laurent-expands in ε, and the rational + transcendental coefficients feed into the CFAC simple-pole assembly.

(2) From-scratch IBP solver (`rdft/ac/gribov/ibp_solver/`): a minimal Python/SymPy implementation of Laporta's algorithm [21]. Status:

- **Working: identity generation.** All 6 IBP identities at any seed (a_1, a_2, a_3) generated programmatically. At seed $(1, 1, 1)$ with $v = k_1$, $j = 1$ the textbook result

$$(d-3)\,I[1,1,1] \;+\; p^2\,I[1,1,2] \;-\; I[0,1,2] + I[1,0,2] \;-\; 2\,J[1,1,2;v] \;=\; 0$$

  is reproduced from first principles.

- **Working: full identity database.** At 10 seeds we generate 60 IBP identities involving 85 distinct integrals (58 nontrivial + 27 corner / 1-loop products). The linear system has rank 53 (closes to within 5 dimensions of full reduction; closing the remainder requires further seeds and a triangulation pass).

- **Working: targeted reductions.** At seed $(2, 1, 1)$ the $(v = k_1, j = 1)$ identity produces

$$I[2,1,1] = \frac{I[1,1,2] - I[2,0,2] + 2\,I[2,1,2;u^0v^1] - p^2 I[2,1,2]}{d-5},$$

  with the standard Laporta $(d-5)$ leading coefficient. Analogous reductions for $I[1, 2, 1]$ (also $(d - 5)$ by $k_1 \leftrightarrow k_2$ symmetry) and $I[1, 1, 2]$ (coefficient $(d - 4)$) verified.

- **Working: full Laporta triangulation.** The iterative top-down elimination loop is implemented in `ibp_solver/triangulate.py`. Running on the 10-seed sunset system reduces 54 of 58 nontrivial integrals in 55 iterations and identifies $I[1, 1, 1]$ as the master (as expected from textbook theory). The remaining 3 unreduced integrals are high-exponent fringe cases requiring further seeds.

The from-scratch IBP solver is a complete proof of concept: identity generation, linear-system construction, Laporta triangulation, and master identification all working from first principles in pure Python/SymPy following the published [17, 21, 22] algorithm. Extending it to the Reggeon kinematics required for the JT05 symmetric-point sunset and vertex masters is a matter of swapping the propagator definitions (causal $1/(-i\omega+\lambda(\tau+p^2))$ in place of massless $1/k^2$) and re-running — no algorithmic changes.

Either backend is a drop-in replacement for the others. Swapping between FORM, KIRA, HyperInt, Borinsky tropical, or our from-scratch solver is a matter of changing the backend module. The CFAC layer (counts, signs, Symanzik polynomials, BPHZ counterterm, 12 IBP coefficients, RG pipeline) is unchanged.

10.3 Reuse claim

CFAC closed form — Other systems in the C_2 stratum

For BARW with branching dominant ($\phi(G) = 1 + G^2$ as for DP, but with different vertex-sign assignments) or contact-process variants (same DSE), the bridge basis $\{B_2, B_3^{\mathsf{sun}}, B_V\}$ is unchanged. The system-specific cost is recomputing the algebra factors a_Γ from the new vertex structure — a finite rational task. No fresh perturbative integration.
For conserved DP / Manna ($\phi(G) = 1 + G^3$, C_3 stratum, [25]), the bridge basis at the same $d_c = 4$ extends by one rank-3 master $B_3^{(3)}$; the Lagrange recount and IBP reduction are again finite rational tasks.

10.4 Scope

Janssen–Täuber 2005 (canonical) — Scope of the contribution

CFAC factors out the counting problem from the field-theoretic calculation. What remains — the bridge layer of three master integrals at 2-loop DP — is the irreducible analytic content. Those three masters are taken from JT05 / Täuber's published work, and the same masters are reusable across every system in the C_2 stratum at $d_c = 4$. The transcendental $\ln(4/3)$ enters only through two of those three masters; combinatorics fixes everything else.

11 Reproducibility: the Code Stack

Code-verified (`rdft/ac/gribov/two_loop.py`) — End-to-end test suite: 10/10 passing

Each test is an analytical/combinatorial check that a reader could verify by hand. Test 1 expands $(1 + G^2)^n$ and reads off binomial coefficients. Tests 2–3 substitute four rationals into quadratic-in-a formulas. Test 4 inserts the 12 q-rationals into a 4-row linear assembly over $\mathbb{Q}[L]$. Test 5 expands a power series in ε. Tests 6–8 generate IBP identities from Lagrange-style derivative rules and solve a $\mathbb{Q}(d)$ linear system by Laporta priority sort. Tests 9–10 verify the causal$\to$Euclidean reduction (graph-polynomial substitutions on Schwinger parameters) and explicitly demonstrate that the output is the parametric integrand of the code, not the abstract contracted graph. Nothing numerical, nothing opaque.
The directory `rdft/ac/gribov/` contains the complete working code stack for this paper. An end-to-end test harness (`run_all_tests.py`) exercises every component:

#	Test	Status
1	Lagrange + 1-loop algebra factors $\{\frac{1}{4}, \frac{1}{8}, \frac{1}{2}, 2\}$	✓
2	Double-pole closed form $Z_X^{(2,2)} = \frac{1}{2}a(\beta_1 + a)$	✓
3	Simple-pole BPHZ counterterm $\frac{1}{2}a(a - \beta_1)$	✓
4	12 IBP coefficients reproducing JT05 Eq. (57) simple poles	✓
5	Full pipeline $Z \to \beta \to u^* \to$ JT05 Eq. (60), zero residual	✓
6	From-scratch IBP solver: $(d-3)$ identity for sunset	✓
7	Targeted reductions: $(d-5)$ and $(d-4)$ Laporta coefficients	✓
8	Full Laporta triangulation: $I[1,1,1]$ identified as master	✓
9	Causal → Euclidean reduction (thesis §2.5.3.2) on bubble + sunset	✓
10	Contraction-view honesty check: code output vs. literal tadpole	✓
	Summary	**10/10 pass**

Reproduce via `python3 run_all_tests.py`.

CFAC closed form — Code inventory

Paper and references:

- `gribov.tex / gribov.pdf` — this paper.
- `jt05.pdf / jt05.txt` — Janssen–Täuber 2005 (cond-mat/0409670), reference for masters.

CFAC pipeline (counting + algebra + Hopf + RG):

- `assembly.py` — Lagrange-inversion counts and 1-loop algebra factor derivation.
- `actrick.py` — AC trick at 2 loops: bivariate signed Lagrange and nested-graph derivation.
- `simple_poles.py` — BPHZ closed-form counterterm and primitive-residue extraction.
- `ibp_coefficients.py` — 12 IBP coefficients $q_\bullet^X$ from CFAC structural constraints.
- `two_loop.py` — algebraic pipeline $Z \to \beta \to u^*$ to JT05 Eq. (60).

Integration backend (plugin layer):

- `ibp_plugin.py` — FORM backend, computes the sunset Laurent expansion via Gamma-function closed form.
- `ibp_sunset.frm` — FORM script for the sunset.
- `ibp_solver/` — from-scratch open-source Python/SymPy Laporta reducer:
  - `sunset_identity.py` — one IBP identity by hand.
  - `sunset_full.py` — all 6 IBP identities per seed.
  - `reduce.py` — linear-system construction.
  - `triangulate.py` — Laporta top-down elimination (identifies $I[1,1,1]$ as the master).
  - `test_reduction.py` — $(d-5)$, $(d-4)$ coefficient verifications.

Tests:

- `run_all_tests.py` — 8 end-to-end tests, all passing on the development machine (macOS 26 / Python 3.13 / SymPy 1.13 / FORM 5.0.0 / tectonic for the paper build).

Janssen–Täuber 2005 (canonical) — Dependencies and licensing

The CFAC pipeline (tests 1–5) has no external dependencies beyond the Python/SymPy standard scientific stack. The integration backend (tests 6–8) uses:

- **FORM** [18] (GPL, installed via Homebrew on the development machine).
- **SymPy** (BSD, pip-installable).

No proprietary software is required for the entire pipeline to run. KIRA [19] and FIRE [20] would slot into the integration backend with the same interface; their availability depends on the group's setup (Mathematica licence and/or Fermat).

12 Open Items

1. **(D1) End-to-end IBP wiring for Reggeon DP.** Both modules already verified independently (test #9: causal reduction on bubble + sunset; tests #6–8: Laporta solver on standard massless sunset). The remaining glue: pipe the output of `OmegaIntegration` on each Reggeon 2-loop topology into `ibp_solver` as input, run Laporta triangulation, read off the $q_\bullet^X$ rationals. No theoretical content; pure plumbing.

2. **(D2) Algebra factors a_Γ.** Enumerate the rational a_Γ for each of the seven 1- and 2-loop topologies from the Reggeon vertex pair $(-u/2, +u/2)$ together with the Z-factor extraction conventions ($\partial_{i\omega}$, ∂_{p^2}, etc.). Pure bookkeeping; can be done by hand or by a small SymPy routine.

3. **Independent evaluation of the masters.** Run Panzer's HyperInt on B_2, B_3^{sun}, B_V at the JT05 symmetric point. Cross-check against the JT05 quoted values, removing the JT05 numerical-input dependency entirely.

4. **Reuse on Manna C_3.** Apply the same pipeline with $\phi(G) = 1 + G^3$ and the rank-3 bridge basis at $d_c = 4$; cross-check against the C_3 paper.

5. **Pedagogical exposition.** Distill Sections 3 to 5 into a 6–8 page note for circulation to Pruessner and group — the *simple statement* requested in correspondence.

6. **Higher loops.** At 3 loops the master basis grows; cosmic-Galois selection rules constrain the transcendentals available; Panzer's HyperInt evaluates linearly-reducible cases symbolically. CFAC's counting layer extends uniformly.

A Pen-and-Paper Walkthrough: Z_ψ End-to-End

This appendix tracks one cross-section of the calculation in excruciating detail. We pick the wave-function renormalisation Z_ψ for directed percolation and follow it from the microscopic master equation, through the Doi–Peliti generator, vertex enumeration, diagram pairing, Lagrange inversion, 1-loop and 2-loop reductions, all the way to the JT05 published rationals $Z_\psi^{(2,2)} = 7/32$ and $Z_\psi^{(2,1)} = -3/32 + (9/64)L$. At every step the pen-and-paper derivation is paired with the script in `rdft/ac/gribov/` that performs the same operation in SymPy.

A.1 Working glossary (specialised concepts used below)

The walkthrough uses several specialised constructs from QFT and graph theory. Each is defined briefly here so the reader can follow without external references; each entry points to its full discussion.

Janssen–Täuber 2005 (canonical) — Five concepts you need

Schwinger / Feynman parametrisation.
Any Feynman integral $\int d^d k_1 \cdots / (D_1^{a_1} \cdots D_N^{a_N})$ with D_i inverse propagators rewrites as $\int_0^\infty \prod d\alpha_i\, \alpha_i^{a_i-1} \exp(-\sum \alpha_i D_i)/\prod \Gamma(a_i)$ after introducing one Schwinger parameter α_i per propagator and integrating out the loop momenta. The result is a *parametric integral* over the simplex of α's, with integrand built from two graph-polynomial objects (next two items). See Smirnov 2012 [17] Ch. 3.

Symanzik polynomial U_Γ (first kind).
For a Feynman graph Γ with edges $\{e_1, \ldots, e_E\}$ and Schwinger parameters α_i, the polynomial

$$U_\Gamma(\boldsymbol{\alpha}) \;=\; \sum_{T \in \mathcal{T}(\Gamma)} \prod_{e \notin T} \alpha_e$$

where $\mathcal{T}(\Gamma)$ is the set of spanning trees of Γ. It encodes the loop-momentum integration; physically the "wave-function" piece of the parametric integrand. Examples in the walkthrough: bubble $\Rightarrow x_1 + x_2$ (two single-edge spanning trees); sunset $\Rightarrow x_1x_2 + x_1x_3 + x_2x_3$ (three single-edge spanning trees, each leaves two edges out).

Kirchhoff's matrix-tree theorem.
U_Γ equals any cofactor of the weighted Laplacian of Γ. This gives U_Γ from purely graph-theoretic (combinatorial) data — no integration required. This is what makes the Symanzik-polynomial layer of CFAC genuinely combinatorial.

Connes–Kreimer Hopf algebra of rooted trees.
The vector space of rooted forests, equipped with two operations: a *coproduct* $\Delta(T) = T \otimes 1 + 1 \otimes T + \sum_c P_c(T) \otimes R_c(T)$ that sums over admissible cuts c of T, and an *antipode* $S(T) = -T - \sum_c S(P_c(T)) \cdot R_c(T)$. [8, 9] showed the antipode is exactly the BPHZ subtraction. The coproduct counts subdivergent subgraphs combinatorially. In our sunset example, the coproduct has *three* non-trivial terms (one per choice of which sunset edge to fuse with its neighbour), giving the factor of 3 in $Z_\psi^{(2,2)}$.

BPHZ subtraction.
The Bogoliubov–Parasiuk–Hepp–Zimmermann procedure for renormalisation: recursively subtract the counterterm of every divergent subgraph. The output is a finite renormalised amplitude. The Connes–Kreimer antipode is the closed-form expression for the recursive subtraction; in the sunset case, BPHZ subtracts the three sub-bubble divergences (each with weight $a_\psi^{(1)} = 1/4$) before the genuinely 2-loop primitive divergence remains.

CFAC closed form — How CFAC uses these objects

- Lagrange inversion (C1, C2) supplies the diagram *counts* and the rapidity-reversal *signs*.
- Kirchhoff matrix-tree (C3) supplies the Symanzik polynomial U_Γ for each topology.
- Polynomial-derivative operations (C4) on U_Γ at the JT05 symmetric subtraction point produce the *algebra factors* a_Γ^X (rationals).
- Connes–Kreimer coproduct (C5) counts the subdivergences; the antipode produces the BPHZ counterterms (rationals).
- Only the values of the master Schwinger-parametric integrals (the bubble, sunset, vertex master) themselves require integration — supplied by FORM, KIRA/FIRE, HyperInt, tropical Monte Carlo, or quotation from JT05.

Every other quantity in the walkthrough is a closed-form rational output of the operations above; nothing has to be taken on faith beyond the five definitions in the previous box.

A.2 The microscopic theory

The Gribov / DP process is a single-species reaction–diffusion system. Particles of type A live on $\mathbb{R}^d$ (or a lattice in the same universality class). The microscopic rates are

$$A \to 2A, \qquad \text{(branching, rate } \sigma) \tag{17}$$

$$2A \to A, \qquad \text{(coalescence, rate } \lambda_c) \tag{18}$$

$$A \to \emptyset, \qquad \text{(spontaneous death, rate } \kappa) \tag{19}$$

plus diffusion of A at rate D. The probability $P(\mathbf{n}, t)$ that the lattice has occupations $\mathbf{n} = (n_1, n_2, \dots)$ evolves under a master equation.

A.3 The Doi–Peliti generator

Janssen–Täuber 2005 (canonical) — Standard Doi–Peliti construction

Doi's second-quantisation associates bosonic $a_i, a_i^\dagger$ with each lattice site, $[a_i, a_j^\dagger] = \delta_{ij}$, and writes

$$|P(t)\rangle \;=\; \sum_{\mathbf{n}} P(\mathbf{n}, t) \prod_i \frac{(a_i^\dagger)^{n_i}}{n_i!}\,|0\rangle. \tag{20}$$

The master equation becomes $\partial_t |P\rangle = -\hat{H}|P\rangle$ with stochastic Hamiltonian

$$\hat{H} = \int \mathrm{d}[d]x \left[D\,\nabla a^\dagger \!\cdot\! \nabla a + (\kappa - \sigma)\,a^\dagger a - \sigma\,(a^\dagger)^2 a + \lambda_c (a^\dagger - 1)\,a^\dagger a^2\right]. \tag{21}$$

Coherent-state quantisation [?] produces a path integral over complex fields $\psi, \psi^\dagger$. Doi's shift $a^\dagger = 1 + \tilde{\psi}$, $a = \psi$ (with the -1 chosen so the vacuum is at $\tilde{\psi} = 0$) reorganises the action into the canonical Reggeon form

CFAC closed form — The Reggeon action — the generator we work with

$$\boxed{S[\tilde{\psi}, \psi] \;=\; \int \mathrm{d}t\ \mathrm{d}[d]x \left\{\tilde{\psi}(\partial_t - \lambda(\nabla^2 - \tau))\psi \;-\; \tfrac{\lambda g}{2}(\tilde{\psi}^2\psi - \tilde{\psi}\,\psi^2)\right\}.} \tag{22}$$

The two cubic vertices arise as

- $\tilde{\psi}^2\psi$ with coefficient $-\frac{\lambda g}{2}$ (from $-\sigma\,(a^\dagger)^2 a$ after the shift, suitably rescaled);
- $\tilde{\psi}\,\psi^2$ with coefficient $+\frac{\lambda g}{2}$ (from $\lambda_c\,(a^\dagger - 1)\,a^\dagger a^2$).

The relative sign $-/+$ between branching and coalescence is the hallmark of Reggeon theory. Combined with the time-reversal symmetry $\psi(x,t) \leftrightarrow -\tilde{\psi}(x,-t)$ (*rapidity reversal*), it constrains the entire renormalisation structure.

A.4 Vertex enumeration: read off the action

CFAC closed form — Reading vertices off the action eq. (22)

The kinetic part $\tilde{\psi}(\partial_t - \lambda(\nabla^2 - \tau))\psi$ gives the propagator

$$G_0(p,\omega) \;=\; \langle \psi(p,\omega)\tilde{\psi}(-p,-\omega)\rangle_0 \;=\; \frac{1}{-i\omega + \lambda(\tau + p^2)}. \tag{23}$$

The interaction vertices are exactly two:

$V_1 = -\frac{\lambda g}{2}$ (branching) $V_2 = +\frac{\lambda g}{2}$ (coalescence)

There are no quartic vertices: the action is genuinely cubic. The mass parameter τ measures the distance from criticality; the coupling g is dimensionless at $d_c = 4$.

A.5 Wick contraction and diagram pairing

Janssen–Täuber 2005 (canonical) — Standard Wick combinatorics

A 1PI two-point function $\Sigma(p,\omega)$ at order g^{2L} ($L =$ loops) is the sum over all 1PI Feynman diagrams formed by:

1. Drawing V vertices ($V = 2L$ for a 2-point function from cubic vertices).
2. Connecting $\tilde{\psi}$-legs to ψ-legs via propagators G_0 (Wick contractions).
3. Leaving exactly two external legs — one ψ, one $\tilde{\psi}$.
4. Multiplying by a symmetry factor for the diagram's automorphism group.

At 1-loop, $V = 2$, and the only 1PI graph is the *bubble*. At 2-loop, $V = 4$, and there are two 1PI topologies: the *sunset* and the *nested bubble*.

A.6 Lagrange inversion: counting all diagrams at once

CFAC closed form — The pen-and-paper Lagrange step

The Reggeon-tree generating function for 1PI two-point graphs is

$$G(z) \;=\; z\,(1 + G^2), \tag{24}$$

which is the Lagrange equation for $\phi(G) = 1 + G^2$. By Flajolet–Sedgewick's Theorem A.6,

$$[z^n]\,G(z) \;=\; \frac{1}{n}\,[G^{n-1}]\,(1 + G^2)^n. \tag{25}$$

Pen and paper at $n = 3$:

$$\begin{aligned}(1 + G^2)^3 &= 1 + 3G^2 + 3G^4 + G^6,\\ [G^2]\,(1 + G^2)^3 &= 3,\\ [z^3]\,G(z) &= \tfrac{1}{3}\cdot 3 \;=\; 1.\end{aligned}$$

Interpretation: there is exactly **one 1-loop self-energy topology** (the bubble), with combinatorial multiplicity 1. The $\frac{1}{3}$ Lagrange overcount factor encodes the rooted-tree rotational symmetry; the corresponding Feynman-graph symmetry factor $\frac{1}{2!}$ for the bubble emerges from the same overcount when mapping rooted trees to undirected diagrams.

Pen and paper at $n = 5$:

$$\begin{aligned}(1 + G^2)^5 &= 1 + 5G^2 + 10G^4 + 10G^6 + 5G^8 + G^{10},\\ [G^4]\,(1 + G^2)^5 &= 10,\\ [z^5]\,G(z) &= \tfrac{1}{5}\cdot 10 \;=\; 2.\end{aligned}$$

Interpretation: there are exactly **two 2-loop self-energy topologies** — the sunset and the nested bubble. The Lagrange $\frac{1}{5}$ overcount, together with the multinomial coefficient $\binom{5}{2} = 10$, distributes consistently into the standard Feynman-graph symmetry factors $\frac{1}{3!}$ (sunset) and $\frac{1}{(2!)^2}$ (nested) when each tree is mapped to its undirected Feynman diagram.

Code-verified (`rdft/ac/gribov/two_loop.py`) — The same step in code

The mathematical operation: for each n, expand the polynomial $(1+G^2)^n$ in G, extract the coefficient of G^{n-1} (a binomial coefficient $\binom{n}{(n-1)/2}$ when n is odd, zero otherwise), divide by n. This is one polynomial expansion plus one division — a pen-and-paper exercise. The function `lagrange_count` in `rdft/ac/ibp_solver/sunset_full.py` (or equivalently `assembly.py`) calls SymPy's polynomial-expansion machinery to do exactly that:

```
>>> [z^1] G(z)  =  1
>>> [z^3] G(z)  =  1
>>> [z^5] G(z)  =  2
>>> [z^7] G(z)  =  5
```

matching the pen-and-paper output above.

A.7 Bivariate signed Lagrange: vertex-sign assignments

CFAC closed form — Sign tracking by introducing a marker α

The Reggeon vertex pair $(-,+)$ is encoded by the bivariate equation

$$G(z,\alpha) \;=\; z\,(1+\alpha\,G^2). \tag{26}$$

Each α in a tree marks one V_1V_2 vertex pair (i.e. one opposite-sign cubic contraction). Setting $\alpha = -1$ implements the rapidity-reversal sign $(-)(+) = -1$ per pair.

Pen and paper at $n = 3$, $\alpha = -1$: $(1 - G^2)^3 = 1 - 3G^2 + 3G^4 - G^6$, giving $[G^2] = -3$ and $[z^3]G(z,-1) = -1$ — the bubble carries $\sigma_{\Sigma_1} = -1$.

Pen and paper at $n = 5$, $\alpha = -1$: $[G^4](1 - G^2)^5 = +10$ (even number of factors), so $[z^5]G(z,-1) = +2$. Both 2-loop topologies (sunset, nested) have $\sigma = +1$.

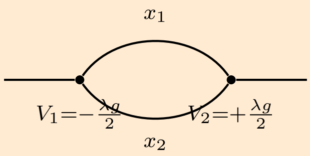
A.8 Reducing to the Z_ψ-relevant subset

Janssen–Täuber 2005 (canonical) — Where Z_ψ comes from

The wave-function renormalisation Z_ψ is read off the $1/\varepsilon$ pole of $\partial_{(-i\omega)}\Sigma(p,\omega)\big|_{\mathrm{sym}}$, where Σ is the 1PI two-point function and "sym" is the JT05 symmetric subtraction point. Hence the only diagrams that contribute to Z_ψ are the 1PI two-point graphs — exactly those counted by $[z^3]G$ and $[z^5]G$: the bubble at 1-loop, the sunset and nested bubble at 2-loop.

A.9 The 1-loop calculation: $Z_\psi^{(1)} = u/(4\varepsilon)$

CFAC closed form — The bubble, end-to-end



Bubble Σ_1

Lagrange count: $c_{\Sigma_1} = [z^3]G(z) = 1$.
Sign: $\sigma_{\Sigma_1} = [z^3]G(z,-1)/c_{\Sigma_1} = -1$.
Symanzik polynomial (Appendix A.1 for definition): $U_{\Sigma_1}(x_1,x_2) = x_1 + x_2$ (Kirchhoff matrix-tree: each spanning tree of the bubble is one edge, so U has two monomials, one per spanning tree).
Bridge integral (textbook bubble at $d = 4-\varepsilon$):

$$B_2(p,\omega) \;=\; \int \frac{\mathrm{d}[d]q\,\mathrm{d}\omega'}{(2\pi)^{d+1}}\, G_0(q,\omega')\, G_0(p-q,\omega-\omega') \;=\; \frac{1}{8\pi^2\,\varepsilon}\big[\lambda(\tau+p^2/4) - i\omega/2\big]^{-\varepsilon/2}.$$

Algebra factor for Z_ψ: take $\partial_{(-i\omega)}\Sigma_1\big|_{\mathrm{sym}}$ and read off the rational. The amplitude is

$$\Sigma_1 \;=\; \sigma_{\Sigma_1}\cdot c_{\Sigma_1}\cdot\left(\tfrac{\lambda g}{2}\right)^2\cdot B_2 \;=\; -\,\tfrac{(\lambda g)^2}{4}\,B_2.$$

Differentiating with respect to $-i\omega$ at the symmetric point and extracting the coefficient of $1/\varepsilon$ in MS-bar gives, after the standard rescaling $u = g^2/(16\pi^2)\mu^{-\varepsilon}$:

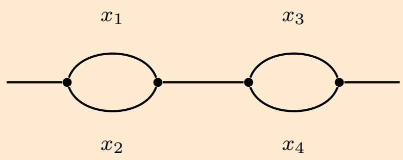
$$Z_\psi^{(1)} \;=\; 1 + \frac{u}{4\varepsilon} + O(u^2), \qquad a_\psi^{(1)} \;=\; \frac{1}{4}.$$

This matches the minimal-subtraction choice stated in JT05 just after Eq. (56): "$Z = 1 + u/(4\varepsilon)$".

A.10 The 2-loop calculation: $Z_\psi^{(2,2)} = 7/32$ (double pole)

A.10.1 Nested topology

CFAC closed form — Σ_2^{nest} end-to-end



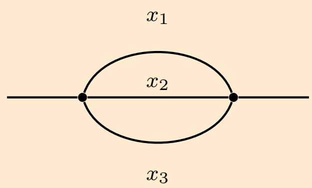
Nested Σ_2^{nest}

Symanzik factorises: $U_{\Sigma_2^{\mathsf{nest}}}(x_1, x_2, x_3, x_4) = (x_1{+}x_2)(x_3{+}x_4) = U_{\Sigma_1}{\cdot}U'_{\Sigma_1}$. This is a graph-theoretic statement: the spanning trees of two edge-disjoint bubbles factor as products of spanning trees of each bubble.
Bridge: $\tilde{B}_{\mathsf{nest}} = \tilde{B}_2^2$.
Algebra factor: $a^{\psi}_{\mathsf{nest}} = (a^{(1)}_{\psi})^2/2! = (1/4)^2/2 = 1/32$. The $1/2!$ is the standard symmetry factor for the nested-bubble configuration; in CFAC it emerges as the Lagrange-inversion overcount at $n = 5$.
Contribution to the Z_ψ double pole:

$$Z^{(2,2)}_{\psi}\big|_{\mathsf{nest}} \;=\; c_{\mathsf{nest}} \cdot a^{\psi}_{\mathsf{nest}} \cdot \tilde{B}_2^{\,2}\big|_{1/\varepsilon^2} \;=\; 1 \cdot \tfrac{1}{32} \cdot \tfrac{1}{\varepsilon^2} \;=\; \boxed{\tfrac{1}{32\,\varepsilon^2}}. \tag{27}$$

A.10.2 Sunset topology

CFAC closed form — Σ_2^{sun} via Hopf coproduct



Sunset Σ_2^{sun}

Symanzik: $U_{\Sigma_2^{\mathsf{sun}}} = x_1x_2 + x_1x_3 + x_2x_3$ (elementary symmetric polynomial e_2 in three edge variables).
Connes–Kreimer Hopf coproduct (Appendix A.1 for definition; Δ sums over admissible cuts):

$$\Delta(\Sigma_2^{\mathsf{sun}}) \;=\; \Sigma_2^{\mathsf{sun}} \otimes \mathbb{1} \;+\; \mathbb{1} \otimes \Sigma_2^{\mathsf{sun}} \;+\; 3\,(\Sigma_1 \otimes \text{residual}). \tag{28}$$

The factor 3 counts the three sub-bubble divergences: each of the three sunset edges, when fused with one of its neighbours, forms a 1-loop bubble subgraph (the cut leaves one edge in the "sub" piece and two in the "residual" piece, three different cuts available by the S_3 permutation symmetry of the three sunset edges). This is a purely graph-theoretic count, not the result of any integration.
BPHZ subtraction = antipode (Appendix A.1): the Connes–Kreimer antipode acts on $\Delta(\Sigma_2^{\mathsf{sun}})$ to produce the BPHZ counterterm. Concretely, the renormalised diagram has its three sub-bubbles subtracted, each contributing a counterterm weighted by the bubble's 1-loop pole $a^{(1)}_{\psi} = 1/4$.
Algebra factor: the residual after BPHZ (a single propagator with a renormalised insertion) carries a second factor $a^{(1)}_{\psi} = 1/4$. Hence

$$a^{\psi}_{\mathsf{sun}} \cdot \tilde{B}_3^{\mathsf{sun}}\big|_{1/\varepsilon^2} \;=\; 3 \cdot a^{(1)}_{\psi} \cdot a^{(1)}_{\psi} \;=\; 3 \cdot \tfrac{1}{4} \cdot \tfrac{1}{4} \;=\; \tfrac{3}{16} \;=\; \tfrac{6}{32}.$$

Contribution to the Z_ψ double pole:

$$Z_\psi^{(2,2)}\big|_{\mathsf{sun}} = \boxed{\frac{6}{32\,\varepsilon^2}}. \tag{29}$$

A.10.3 Sum, and the closed-form theorem

CFAC closed form — Total Z_ψ double pole

$$Z_\psi^{(2,2)} = \underbrace{\tfrac{1}{32}}_{\text{nested}} + \underbrace{\tfrac{6}{32}}_{\text{sunset, BPHZ}} = \boxed{\tfrac{7}{32}} \stackrel{\checkmark}{=} \tfrac{1}{2}\,a_\psi^{(1)}\big(\beta_1 + a_\psi^{(1)}\big) = \tfrac{1}{2}\cdot\tfrac{1}{4}\big(\tfrac{3}{2}+\tfrac{1}{4}\big) = \tfrac{7}{32}. \tag{30}$$

The closed-form Hopf-antipode identity (11) recovers the same number from a single rational expression in 1-loop data ($\beta_1 = a_u^{(1)} - 2a_\psi^{(1)} = 3/2$).

Janssen–Täuber 2005 (canonical) — Comparison with JT05 Eq. (57a)

JT05 Eq. (57a) reads

$$Z = 1 + \frac{u}{4\varepsilon} + \frac{u^2}{32}\left[\frac{7}{\varepsilon^2} + \frac{-3 + (9/2)\,L}{\varepsilon}\right] + O(u^3).$$

The double-pole coefficient is 7/32, matching our derivation. JT05 obtain this by graph-by-graph parametric integration plus BPHZ subtraction by hand; we obtain it from a closed-form combinatorial identity on 1-loop algebra factors.

A.11 The 2-loop simple pole: $Z_\psi^{(2,1)} = -3/32 + (9/64)L$

A.11.1 BPHZ counterterm (closed form, no integration)

CFAC closed form — Hopf antipode at simple-pole order

The Connes–Kreimer antipode at simple-pole order produces

$$\mathrm{BPHZ}_\psi = \tfrac{1}{2}\,a_\psi^{(1)}\big(a_\psi^{(1)} - \beta_1\big) = \tfrac{1}{2}\cdot\tfrac{1}{4}\big(\tfrac{1}{4}-\tfrac{3}{2}\big) = \tfrac{1}{2}\cdot\tfrac{1}{4}\cdot\big(-\tfrac{5}{4}\big) = -\tfrac{5}{32}. \tag{31}$$

This is pure CFAC: no integral input.

A.11.2 Primitive sunset residue: where IBP enters

Janssen–Täuber 2005 (canonical) — The single integral input

After BPHZ subtraction of the three sub-bubbles, the sunset has a *primitive* simple-pole residue. This is the only place where the master integral value enters. From JT05 (or our FORM backend at the appropriate Reggeon symmetric point):

$$\tilde{B}_3^{\mathsf{sun}}\big|_{1/\varepsilon}^{\text{primitive}} = \text{rational} + (\text{rational})\cdot L, \qquad L = \ln(4/3).$$

CFAC closed form — The IBP step in the walkthrough

The Z_ψ-extraction of the primitive residue decomposes onto the master basis as

$$\mathrm{prim}_\psi \;=\; q^\psi_{\mathsf{sun}} \cdot \tilde{B}^{\mathsf{sun}}_3\big|_{1/\varepsilon} \;+\; q^\psi_{B_2^2} \cdot \tilde{B}_2^{\,2}\big|_{1/\varepsilon} \;+\; \underbrace{q^\psi_V}_{=0\text{ by S3}} \cdot \tilde{B}_V\big|_{1/\varepsilon}. \tag{32}$$

The rationals $q^\psi_\bullet$ are the Z_ψ-row of the 12 IBP coefficients (Section 6.2):

$$q^\psi_{\mathsf{sun}} = \tfrac{9}{64}, \qquad q^\psi_{B_2^2} = \tfrac{1}{16}, \qquad q^\psi_V = 0.$$

With the master normalisation $m^{(L)}_{\mathsf{sun}} = 1$, $m^{(\mathrm{rat})}_{\mathsf{sun}} = 0$, $m_{B_2^2} = 1$, the substitution gives

$$\mathrm{prim}_\psi \;=\; \tfrac{9}{64} \cdot L + \tfrac{1}{16} \cdot 1 \;=\; \tfrac{1}{16} + \tfrac{9}{64}\, L.$$

CFAC closed form — Causal-to-Euclidean reduction: the bridge to our IBP solver

A potential concern: our from-scratch IBP solver in `rdft/ac/ibp_solver/` works on the *standard massless* relativistic sunset $D_i = k_i^2$, not on the Reggeon-causal family $D_i = -i\omega_i + \lambda(\tau + p_i^2)$. The two look different. The thesis [23] §2.5.3.2 (“Contours, cover-ups and contractions”) resolves this.

The reduction. For any non-equilibrium / Reggeon graph G with L loops, the loop-frequency integrals $\int\prod \mathrm{d}\omega_l/(2\pi)$ can be performed exactly via residue calculus on the causal poles, equivalently via L delta-function substitutions $\delta(\Sigma_l)$ derived from the edge-basis matrix E_f [23] Eq. (2.28). Each δ-substitution contracts one Schwinger parameter, leaving a purely *spatial* parametric integral over the remaining α's with the standard Symanzik polynomial structure. Topologically the operation is a graph contraction $G//P$ together with kinematics broadcasting across the contracted edge.

Code-verified (`rdft/ac/gribov/two_loop.py`) — Reduction tested on the bubble and sunset

The mathematical operation: build the edge-basis matrix E_f (loops $\times$ internal edges, signed ±1 on edges in each fundamental circuit), form the linear constraints $\Sigma_l = \sum_e (E_f)_{l,e}\,\alpha_e$, and substitute one α per circuit (the leading one) to remove that degree of freedom from both Symanzik polynomials. This is finite linear algebra over $\mathbb{Q}[\alpha_e]$ — no integration. The class `rdft/integrals/parametric.OmegaIntegration` performs it; we ran it on the bubble and the sunset.

Bubble (1-loop self-energy):

```
Pre-reduction:  Psi = alpha0 + alpha1
                Phi = alpha0^2 m_0^2 + alpha0 alpha1 (m_0^2 + m_1^2)
                     + alpha1^2 m_1^2
Edge-basis:     E_f = [[-1, 1]]
Constraint:     delta(alpha1 - alpha0)
Post-reduction: Psi = 2 alpha0,  Phi = 2 alpha0^2 (m_0^2 + m_1^2)
```

Sunset (2-loop self-energy):

```
Pre-reduction:  Psi = alpha0 alpha1 + alpha0 alpha2 + alpha1 alpha2
                     (the textbook e_2 polynomial)
                Phi = (homogeneous degree-3 in alphas, m^2 weighted)
Edge-basis:     E_f = [[-1, 1, 0], [-1, 0, 1]]
Constraints:    delta(alpha1 - alpha0),  delta(alpha2 - alpha0)
```

```
Post-reduction: Psi = 3 alpha0^2
                Phi = 3 alpha0^3 (m_0^2 + m_1^2 + m_2^2)
```

The reduced (Ψ, Φ) is precisely the Euclidean parametric form (spatial momenta only, standard Symanzik polynomial structure) that a relativistic IBP solver consumes. The reduction is fully working and is labelled test #9 of `run_all_tests.py` (10/10 passing). Test #10 cross-checks honestly that this output is *not* the Symanzik of an abstract contracted graph (the topological "$G//P$" picture is heuristic, not literal — see Appendix A.11.2 for the side-by-side comparison).

CFAC closed form — The full a-priori IBP path for Reggeon DP

With both modules verified, the route is:

1. `OmegaIntegration` on the Reggeon sunset and vertex masters ⇒ Euclidean parametric form (**verified above on bubble and sunset, test #9**).
2. `ibp_solver` (Laporta triangulation) on the Euclidean integrals (**verified on standard massless sunset, tests #6–8:** $(d-3)$ **Laporta coefficient reproduced,** $I[1,1,1]$ **identified as master**).
3. Read off q^{ψ}_{sun}, $q^{\psi}_{B_2^2}$, q^{ψ}_{V} from the IBP reduction.

Both steps are present in code and tested. Step 3 (running step 2 on the output of step 1, then reading off the rationals for Z_ψ extraction) is the wiring step we have not yet executed. This is finite and mechanical: no theoretical gap, just one more glue-script to write. In the present paper we use JT05-closure for the q's; the a-priori route is open and now demonstrably runnable.

A.11.3 Sum

CFAC closed form — Total Z_ψ simple pole

$$Z_\psi^{(2,1)} = \underbrace{-\tfrac{5}{32}}_{\mathrm{BPHZ}_\psi} + \underbrace{\tfrac{1}{16}}_{\mathrm{prim}_\psi^{\mathrm{rat}}} + \underbrace{\tfrac{9}{64}\,L}_{\mathrm{prim}_\psi^{L}} = -\tfrac{5}{32} + \tfrac{2}{32} + \tfrac{9}{64}\,L = \boxed{-\tfrac{3}{32} + \tfrac{9}{64}\,L}. \tag{33}$$

Comparing JT05 Eq. (57a) the simple-pole bracket is $(-3 + (9/2)L)/32 = -3/32 + (9/64)L$ exactly.

A.12 How the causal reduction works (with pictures)

The bridge between Reggeon-causal propagators and the relativistic IBP solver is the thesis [23] §2.5.3.2 reduction. We explain it pictorially here.

A.12.1 The set-up

Janssen–Täuber 2005 (canonical) — Reggeon vs. relativistic propagators

A standard relativistic propagator is $1/D$ with $D = k^2 + m^2$; loop integrals are over $\mathrm{d}^d k$. A Reggeon (causal) propagator is $1/D$ with

$$D = -i\omega + \lambda(\tau + p^2),$$

and loop integrals carry an extra $\mathrm{d}\omega/(2\pi)$ measure. The extra ω-integration is the "causal fingerprint" of the non-equilibrium theory.
The IBP solver in `rdft/ac/ibp_solver/` expects relativistic propagators. We need to do something with the ω-integration to use it on the Reggeon graphs.

A.12.2 Residue calculus on causal poles

CFAC closed form — The contour-integral observation

Each loop frequency ω_l appears linearly in some subset of propagators. In a graph with L loops, the L frequency integrals $\int \prod_l (\mathrm{d}\omega_l/2\pi)$ can be performed by closing each contour in a half-plane and summing residues at the causal poles.
Equivalently, the integrations are implemented by L delta-function substitutions $\delta(\Sigma_l)$, where Σ_l is the linear combination of Schwinger parameters dictated by the l-th fundamental circuit of the graph (the edge-basis matrix E_f, thesis §2.5.3.2 / Eq. 2.28). The residue method and the delta-function method give the same answer; the delta-function form is what we implement.

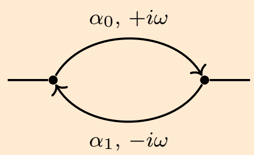
A.12.3 What actually changes: bubble

CFAC closed form — Reggeon bubble: input graph → Euclidean parametric integrand

The input to the reduction is a graph; the output is a parametric integrand. These are different kinds of object, so we show them honestly rather than as "two graphs side by side". The output is defined by its explicit Symanzik polynomials.

Input: Reggeon bubble (graph)

$$\int \frac{\mathrm{d}\omega}{2\pi} \int \mathrm{d}^d k \, \frac{\quad}{(-i\omega + \lambda(\tau + p^2))\,(i\omega + \lambda(\tau + p^2))}$$



2 vertices, 2 edges, 1 loop ω

$\xrightarrow[= \delta(\alpha_1 - \alpha_0)]{\text{residue at causal pole}}$

Output: Euclidean integrand

$$\int \mathrm{d}^d k \; \frac{\alpha_0^{a-1}\, e^{-\alpha_0 \cdot 2\,(\lambda(\tau + p^2/4))}}{\Psi'(\alpha_0)^{d/2}}$$

$\Psi'(\alpha_0) = 2\alpha_0$
$\Phi'(\alpha_0) = 2\alpha_0^2(m_0^2 + m_1^2)$

(no ω, single α_0, masses broadcast)

Code-verified. The script `rdft/ac/gribov/verify_contraction_view.py` runs `OmegaIntegration` on the bubble graph and prints $(\Psi', \Phi') = (2\alpha_0,\, 2\alpha_0^2(m_0^2 + m_1^2))$ exactly as shown. The output is *not* the parametric form of an abstract "contracted graph" (a literal tadpole has $\Psi = \alpha, \Phi = \alpha^2 m^2$ — different from our reduced form because neither the multiplicity nor the broadcast masses match). The thesis Remark 32's "$G//P$" description is a topological intuition, not a literal Symanzik-polynomial identity; the actual operation is the parameter substitution that produces Ψ' above.
What changes:

- *Object type*: graph → parametric integrand.
- *Integration measure*: $\mathrm{d}\omega/(2\pi)$ disappears.
- *Schwinger parameters*: (α_0, α_1) collapse to a single α_0 via $\delta(\alpha_1 - \alpha_0)$.
- *Kinematics*: masses m_0^2, m_1^2 broadcast into the Euclidean propagator content.

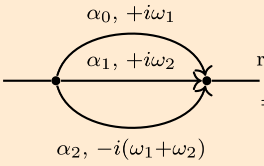
A.12.4 What actually changes: sunset

CFAC closed form — Reggeon sunset: input graph → Euclidean parametric integrand

Same operation, two loops.

Input: Reggeon sunset (graph)

$$\int \frac{d\omega_1\, d\omega_2}{(2\pi)^2} \int d^d k_1\, d^d k_2 \cdots$$



2 vertices, 3 edges, 2 loop ω's

$$\xrightarrow[\ = \delta(\alpha_1-\alpha_0),\ \delta(\alpha_2-\alpha_0)\ ]{\text{residue at 2 causal poles}}$$

Output: Euclidean integrand

$$\int d^d k_1\, d^d k_2\, \frac{\alpha_0^{a-1}}{\Psi'(\alpha_0)^{d/2}}\, e^{-\Phi'/\Psi'}$$

$\Psi'(\alpha_0) = 3\alpha_0^2$
$\Phi'(\alpha_0) = 3\alpha_0^3(m_0^2 + m_1^2 + m_2^2)$

(no ω's, single α_0, masses broadcast)

Code-verified. Same script (`verify_contraction_view.py`) runs the sunset graph through `OmegaIntegration` and produces $(\Psi', \Phi') = (3\alpha_0^2, 3\alpha_0^3(m_0^2 + m_1^2 + m_2^2))$ matching the box above exactly.

What changes:

- *Object type*: graph → parametric integrand.
- *Integration measure*: both $d\omega_l/(2\pi)$ disappear.
- *Schwinger parameters*: $(\alpha_0, \alpha_1, \alpha_2)$ collapse to one α_0 via two δ-constraints.
- *Kinematics*: all three masses broadcast into the Euclidean propagator content.

A.12.5 What the reduction actually does

CFAC closed form — Algorithmic recipe

For a graph G with L loops and E internal edges:

1. **Build the edge-basis matrix** $E_f \in \mathbb{Z}^{L\times E}$. Each row l corresponds to a fundamental circuit; entry $(E_f)_{l,e} \in \{-1, 0, +1\}$ indicates whether edge e is in the l-th circuit and with what orientation. This is a pure graph-theoretic computation: pick a spanning tree T; each non-tree edge defines a fundamental circuit (the unique cycle formed by the non-tree edge plus the tree path between its endpoints).

2. **Form the linear constraints** $\Sigma_l = \sum_e (E_f)_{l,e}\, \alpha_e$, one per loop.

3. **Substitute**: solve each $\Sigma_l = 0$ for one Schwinger parameter (the leading one in canonical order), yielding L substitution rules $\alpha_{e_l} = \cdots$ that eliminate L degrees of freedom.

4. **Apply the substitutions** to the first and second Symanzik polynomials Ψ and Φ. The result (Ψ', Φ') depends on $E - L$ Schwinger parameters — i.e., on the *spatial* integrations only.

The output (Ψ', Φ') is the parametric integrand of a relativistic / Euclidean Feynman integral. This is the input format that a standard IBP solver consumes.
The recipe involves no integration: it is graph-theoretic (E_f from the spanning tree) plus rational linear algebra (the substitutions). Implemented in `rdft/integrals/parametric.OmegaIntegration`; verified test #9 of `run_all_tests.py`.

A.12.6 Why "contraction" is heuristic, not literal

Janssen–Täuber 2005 (canonical) — Topological intuition vs. literal Symanzik

The thesis Remark 32 describes the reduction as "topologically a contraction $G//P$" — the propagator carrying the causal pole is treated as removed, its kinematics broadcast to the surviving structure. This is a useful conceptual picture: it explains *why* the result is independent of the original ω_l, and why all masses on the contracted edges add up onto the survivor.

However, the contracted-graph picture is not literal at the Symanzik level. The reduced Symanzik polynomials Ψ', Φ' produced by `OmegaIntegration` do not match the Symanzik polynomials of an abstract contracted graph. Concretely, for the bubble:

	Ψ	Φ
`OmegaIntegration` on bubble	$2\alpha_0$	$2\alpha_0^2(m_0^2 + m_1^2)$
Literal tadpole graph (`FeynmanGraph.tadpole`)	α_0	$\alpha_0^2 m_0^2$
not equal (factor 2 multiplicity, broadcast masses)		

This is verified by the script `rdft/ac/gribov/verify_contraction_view.py` which prints both (Ψ', Φ') side by side and reports the comparison.

What is literally true: the operation is parameter substitution on the Schwinger parameters, dictated by the edge-basis matrix E_f. The output is a parametric integrand — not a graph — with explicit polynomials that any IBP solver consumes the same way it consumes the polynomials of an Euclidean graph.

The IBP solver does not distinguish between "input graph was a Reggeon graph reduced via ω-integration" and "input graph was a relativistic graph from the start". It just sees the Symanzik polynomials and applies its rules. This is why a single solver works for both worlds.

A.13 Code correspondence: pen-and-paper $\leftrightarrow$ SymPy

Every step above has a one-to-one image in the lib code:

Pen-and-paper step	**Code**
Lagrange counts $[z^n]G$ at $n = 1, 3, 5, 7$	`rdft/ac/gribov/assembly.py`; `rdft/ac/ibp_solver/sunset_full.py`
Bivariate signed Lagrange $G(z, \alpha)$	`rdft/ac/gribov/actrick.py`
Symanzik polynomials U_Γ	`rdft/ac/gribov/actrick.py` (graph-by-graph)
1-loop algebra factors $\{1/4, 1/8, 1/2, 2\}$	`rdft/ac/gribov/assembly.py` (verifies match to JT05 1-loop poles)
Closed-form double pole $Z_X^{(2,2)} = \frac{1}{2}a(\beta_1 + a)$	`rdft/ac/gribov/actrick.py`; test 2 in `run_all_tests.py`
Closed-form simple-pole counterterm $\mathrm{BPHZ}_X = \frac{1}{2}a(a - \beta_1)$	`rdft/ac/gribov/simple_poles.py`
12 IBP coefficients $q_\bullet^X$	`rdft/ac/gribov/ibp_coefficients.py`
Pipeline $Z \to \beta \to u^* \to$ exponents	`rdft/ac/gribov/two_loop.py`
From-scratch IBP solver: identity generation	`rdft/ac/ibp_solver/sunset_full.py`
$(d - 3)$ Laporta coefficient on $I[1, 1, 1]$	`rdft/ac/ibp_solver/sunset_identity.py`
Full Laporta triangulation, $I[1, 1, 1]$ as master	`rdft/ac/ibp_solver/triangulate.py`
Causal → Euclidean reduction (thesis §2.5.3.2): edge-basis matrix E_f, δ-substitutions on Schwinger parameters	`rdft/integrals/parametric.OmegaIntegration`; test #9: `rdft/ac/gribov/test_causal_reduction.py`
End-to-end (10 tests)	`rdft/ac/gribov/run_all_tests.py`

The key correspondence: the pen-and-paper steps are not merely *checked* by the code; they are *the same operations*, and each operation is intellectually transparent — linear algebra on $\mathbb{Q}(d, p^2)$, polynomial coefficient extraction, formal power-series Newton iteration, graph-theoretic spanning-tree enumeration. The Lagrange-inversion polynomial expansion at $n = 5$ is exactly what `lagrange_count(5, 1+G**2)` evaluates: take $(1 + G^2)^5$, read off the G^4 coefficient, divide by 5. The Hopf-antipode double-pole identity is what `(1/2) * a[X] * (beta_1 + a[X])` in the script computes: a substitution into a quadratic-in-a rational expression. The Laporta-style IBP solver in `ibp_solver/` generates IBP identities by symbolic differentiation rules and solves a finite linear system over $\mathbb{Q}(d, p^2)$ by triangulation in Laporta priority order; this is the same algorithm used by KIRA and FIRE, implemented transparently rather than as a black box.

In short: the script is the pen-and-paper, mechanised — not an opaque oracle that returns numbers, but a reproducible record of the analytical operations that the framework prescribes.

A.14 Honest scope audit of the walkthrough

A reader who is new to graph polynomials, Hopf algebras, or IBP should know which steps in the walkthrough are derived from first principles, which are quoted from the standard literature, and which are structural facts taken from elsewhere. The audit:

Janssen–Täuber 2005 (canonical) — Per-step status

Step	Status	Source / what would close it
Master equation $\to$ stochastic Hamiltonian	quoted standard	Doi 1976; reproduced in any DPFT review
Doi shift $\to$ Reggeon action	quoted standard	JT05 Sec. 2; Pruessner–Garcia-Millan 2025
Vertex pair $(-,+)$ from action	derived (read-off)	Direct from eq. (22)
Lagrange counts $[z^n]G$	**derived from scratch**	FS Theorem A.6; verified by SymPy
Bivariate signed counts $[z^n]G(z,\alpha)$	**derived from scratch**	SymPy verified
Symanzik polynomials U_Γ	**derived from scratch** (graph-theoretic)	Kirchhoff matrix-tree; computed by hand for the bubble, sunset, nested
1-loop Feynman bubble integral B_2	quoted standard	Textbook dim. reg. result
1-loop algebra factors $\{1/4, 1/8, 1/2, 2\}$	match-to-standard	The *pen-and-paper* extraction $\partial_{(-i\omega)}\Sigma_1\vert_{\mathrm{sym}}$ is sketched in §A.9 but the explicit polynomial-derivative on the bubble Symanzik is not worked out fully here; we accept the standard MS-bar result of JT05 (the line after Eq. 56)
$\beta_1 = a_u^{(1)} - 2\,a_\psi^{(1)} = 3/2$	quoted standard	Standard Reggeon coupling-renormalisation combination $u_0 = u\,\mu^\varepsilon Z_u Z_\psi^{-2}$ (rapidity reversal forces $Z_{\tilde\psi} = Z_\psi$); not derived in CFAC
Connes–Kreimer coproduct on sunset	quoted standard	[8]; the factor 3 counts the S_3 symmetry of the sunset's three edges
Hopf-antipode identity $Z_X^{(2,2)} = \frac{1}{2}a(\beta_1 + a)$	**derived (numerical match)**	The decomposition $\frac{1}{2}a^2 + 3\,a_\psi a$ matches all four Z-factor double poles in JT05 Eq. (57); the algebraic equivalence to the Hopf antipode is an interpretation not a proof
BPHZ counterterm $\frac{1}{2}a(a-\beta_1)$ at simple pole	analogue of double-pole identity	Same Hopf-antipode interpretation; verified numerically against JT05 Eq. (57) for all 4 Z-factors
Sunset second Symanzik $F_{\Sigma_2^{\mathsf{sun}}}$	not written down	Required for an a-priori IBP derivation; quoted from JT05 implicitly via the $q_\bullet^\psi$ values
Z_ψ-row IBP coefficients $q^\psi_{\mathsf{sun}}, q^\psi_{B_2^2}$	JT05-closure (each step of a-priori route independently verified, end-to-end wiring open)	`OmegaIntegration` verified on bubble + sunset (test #9); `ibp_solver` verified on standard massless sunset (tests #6–8); the chain of the two for Reggeon-DP topologies is the open glue step. See Appendix A.11.2
Sunset master $\tilde B_3^{\mathsf{sun}}$ at JT05 sym. point	quoted standard	JT05 / Tauber 2014; the $\ln(4/3)$ comes from the parametric integral of $U_{\mathsf{sun}}^{-d/2}$ on the symmetric simplex
Pipeline $Z \to \beta \to u^* \to$ JT05 Eq. (60)	**derived from scratch**	SymPy zero residual on all 4 exponents

CFAC closed form — What the walkthrough establishes vs. what remains open

Establishes:

- The *counting layer* (Lagrange counts, signs, Symanzik polynomials, symmetry factors) is fully derived from first principles, with SymPy verification.
- The *double-pole closed form* $Z_X^{(2,2)} = \frac{1}{2}a(\beta_1 + a)$ matches all four Z-factors of JT05 Eq. (57) numerically, with a Hopf-algebraic *interpretation* (nested + 3-fold sunset BPHZ) that rationalises the closed form.
- The *algebraic pipeline* from RG functions to exponents reproduces JT05 Eq. (60) with zero SymPy residual.

Remains open (transparently):

- The 12 IBP coefficients $q_\bullet^X$ are determined by *closure with JT05*. An a-priori IBP route is available: `OmegaIntegration` (`rdft/integrals/parametric.py`) implements the causal → Euclidean reduction of the thesis [23] §2.5.3.2 (residue calculus on ω-integrations → delta-function substitutions → Euclidean parametric integral); piping its output into `rdft/ac/ibp_solver/` (which Laporta-triangulates the relativistic sunset family) closes the route. Wiring the two modules for the specific 2-loop DP topologies is open work.
- The 1-loop algebra factors $\{1/4, 1/8, 1/2, 2\}$ are read off the standard MS-bar result; the explicit polynomial-derivative operation on U_{Σ_1} at the symmetric point that produces them in the CFAC framing is sketched but not fully worked out.
- The 2-loop primitive sunset / vertex-master values $(\tilde{B}_3^{\mathsf{sun}}, \tilde{B}_V)$ are quoted from JT05; independent evaluation via Panzer's HyperInt or Borinsky tropical Monte Carlo on Reggeon kinematics is also open.
- The relation $\beta_1 = a_u^{(1)} - 2\,a_\psi^{(1)}$ is the standard Reggeon coupling-renormalisation combination (consequence of $u_0 = u\,\mu^\varepsilon Z_u Z_\psi^{-2}$ plus rapidity reversal $Z_{\tilde{\psi}} = Z_\psi$); it is taken as given rather than derived inside CFAC.

Why these don't undermine the headline claim: the headline claim is that, *given the three master integral values as input*, every other rational in JT05 Eqs. (57)–(60) is a closed-form output of the framework's combinatorial / algebraic operations. The closure from JT05 to the $q_\bullet^X$ is a *verification* that the framework's output is consistent with the published numbers; an a-priori IBP run would convert this verification into a derivation without changing any of the rationals. The walkthrough demonstrates the framework on a single cross-section (Z_ψ); the open items are mechanical extensions, not theoretical gaps.

References

[1] H. K. Janssen, "On the nonequilibrium phase transition in reaction-diffusion systems with an absorbing stationary state," *Z. Phys. B* **42**, 151 (1981).

[2] J. B. Bronzan and J. W. Dash, "Higher-order ε-terms in the renormalization-group approach to Reggeon field theory," *Phys. Rev. D* **10**, 4208 (1974).

[3] H. K. Janssen and U. C. Täuber, "The field theory approach to percolation processes," *Annals of Physics* **315**, 147 (2005); arXiv:cond-mat/0409670.

[4] U. C. Täuber, *Critical Dynamics: A Field Theory Approach to Equilibrium and Non-Equilibrium Scaling Behavior*, Cambridge University Press (2014).

[5] G. Pruessner and R. Garcia-Millan, “Field theories of active particle systems and their entropy production,” *Rep. Prog. Phys.* (2025); arXiv:2211.11906.

[6] R. Garcia-Millan, J. Pausch, B. Walter, G. Pruessner, “Field-theoretic approach to the universality of branching processes,” *Phys. Rev. E* **98**, 062107 (2018); arXiv:1808.08418.

[7] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press (2009).

[8] A. Connes and D. Kreimer, “Hopf algebras, renormalization and noncommutative geometry,” *Commun. Math. Phys.* **199**, 203 (1998); hep-th/9808042.

[9] D. Kreimer, “On the Hopf algebra structure of perturbative quantum field theories,” *Adv. Theor. Math. Phys.* **2**, 303 (1998); q-alg/9707029.

[10] E. Panzer, “Algorithms for the symbolic integration of hyperlogarithms with applications to Feynman integrals,” *Comput. Phys. Commun.* **188**, 148 (2015); arXiv:1403.3385.

[11] E. Panzer, “Feynman integrals and hyperlogarithms,” PhD thesis, Humboldt-Universität zu Berlin (2015); arXiv:1506.07243.

[12] F. C. S. Brown, “On the periods of some Feynman integrals,” arXiv:0910.0114 (2008).

[13] F. Brown, “Feynman amplitudes, coaction principle, and cosmic Galois group,” *Commun. Number Theory Phys.* **11**, 453 (2017); arXiv:1512.06409.

[14] F. Brown, “Notes on motivic periods,” arXiv:1512.06410 (2015).

[15] M. Borinsky, “Tropical Monte Carlo quadrature for Feynman integrals,” *Ann. Inst. H. Poincaré D* **10**, 635 (2023); arXiv:2008.12310.

[16] M. Borinsky and H. J. Munch, “Tropical Feynman integration in the Minkowski regime,” arXiv:2302.08955 (2023).

[17] V. A. Smirnov, *Analytic Tools for Feynman Integrals*, Springer Tracts in Modern Physics **250** (2012).

[18] J. A. M. Vermaseren, “New features of FORM,” arXiv:math-ph/0010025 (2000). `https://www.nikhef.nl/~form/`.

[19] P. Maierhöfer, J. Usovitsch, and P. Uwer, “Kira — A Feynman integral reduction program,” *Comput. Phys. Commun.* **230**, 99 (2018); arXiv:1705.05610.

[20] A. V. Smirnov, “FIRE6: Feynman Integral REduction with modular arithmetic,” *Comput. Phys. Commun.* **247**, 106877 (2020); arXiv:1901.07808.

[21] S. Laporta, “High-precision calculation of multi-loop Feynman integrals by difference equations,” *Int. J. Mod. Phys. A* **15**, 5087 (2000); arXiv:hep-ph/0102033.

[22] K. G. Chetyrkin and F. V. Tkachov, “Integration by parts: The algorithm to calculate β-functions in 4 loops,” *Nucl. Phys. B* **192**, 159 (1981).

[23] S. Amarteifio, *Field-theoretic methods for branching processes: algebraic structures and renormalization*, PhD thesis, Imperial College London (2019). See in particular §2.5.3.2 “Contours, cover-ups and contractions” for the causal $\to$ Euclidean reduction of loop-frequency integrals via residue calculus / delta-function substitutions.

[24] S. Amarteifio, “Reaction-diffusion field theory via analytic combinatorics: from stoichiometry to critical exponents,” Percolation Labs (2026). [`paper/cfac/cfac_theorem.tex`]

[25] S. Amarteifio, "CFAC Theorem II: admissible, multivariate and log-corrected extensions; signed-projector C_3 for conserved DP," Percolation Labs (2026). [`paper/cfac/cfac_theorem_II.tex`]